Recursive self-improvement, gated by evidence

An engine that lets deployed agents improve their own operation — proposing, proving, canarying and shipping their own corrections — measured where the world changes under a long-running agent and outcome learning alone cannot recover

SenseLab Research

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Abstract

Agents that run for months on complex work face a problem no static model solves: the world they operate in changes under them, their tasks never repeat exactly, and every session that acts on what used to be true fails the same way until something in the system changes. Scaling such agents is not a retrieval problem. It is a self-improvement problem: the system must detect that its own knowledge has gone wrong, propose a correction, prove it, release it safely, learn from the result, and do all of this without a person in every loop and without shipping the wrong thing to every agent at once. SenseLab is a recursive self-improvement engine built for that. Outcome evidence changes what agents act on (level 0); a repair agent drafts corrections that must pass a stand-in replay, the customer's own replay and a live canary that shows them no worse than the rule they replace before they ship, reversibly (level 1); the loop's own history feeds its next draft (level 2); and the record it produces trains the next model and the repair prompt itself (level 3). This paper is the first measurement of such a loop. We evaluate the whole engine — never its parts against each other — on a continual benchmark in which the correct action for a task class changes *conditionally* at episode 30 of 100, so that half of the class needs the new rule and half still needs the old one, and no amount of retrying the old action can recover the first attempt. On four held-out seeds, both agent models and two task families (12,800 episodes; analysis frozen before the run) SenseLab takes the new action first in 45 % of moved-half episodes against 25 % for a frontier model alone, 32–36 % for three pgvector baselines and 40 % for a context window carrying every prior note, and holds unchanged work at 100 %. The difference is in SenseLab's favour in 8–11 of 16 cells against every baseline; none of the five paired comparisons survives the Holm correction at 16 cells. Where the context window fits, it wins the half of the class whose rule did not change, by 24 points at three times the cost. Level 1 has not yet added to level 0 inside a 100-episode cell (45 against 43 %), and the record says why: of the 32 fixes the loop proposed in those cells the replay tiers rejected 22 and passed 6, and no canary decided inside a cell. In the confirmatory ten-agent, 200-episode cells the canary arm reads the fix it is graded on (75 % of graded sessions, against 0 before the last two findings), four of six canaries reach the graded floor on both arms, one decides — a failure — and none ships; on the same cells the moved half rises from 37 % in the fifty episodes after the change to 75 % in the last fifty. The paper's main contribution is the record of how a preregistered benchmark found thirteen things about the loop's evidence path — eleven defects, all fixed in production between runs and re-measured on the same protocol — and what each says about building self-improving systems that can be trusted: what counts as a regime shift, how a situation is identified, what a fix must be proved against, who grades the proof, what counts as traffic, which runs a canary can grade, whether the agents can read what the loop wrote, and whether the loop's own tests may be allowed to grade its fixes. The development-seed tables are kept as the record of that development; the confirmatory section is the frozen read.

1 Introduction

The hard problem in scaling agents is not capability but longevity. A model that solves a task today is deployed into an operation that runs for months: the policies it applies are rewritten, the dependencies it pins are audited, the workflows it follows are changed by people who never tell it. Its tasks never repeat exactly, so nothing can be memorised; and the failure that matters is not that it cannot find what it knows but that it acts, confidently, on what *used* to be true — and will do so again on every session that reads the same thing, until something in the system changes. A static model with a static context degrades monotonically in such a world. A person in every loop does not scale.

The path forward is a system that improves its own operation from its own results, and whose improvements compound: recursive self-improvement. It is also the thing everyone is right to be nervous about. A self-modifying system amplifies its own errors as readily as its corrections; a model that drafts fixes from failure evidence will draft wrong ones, and a wrong fix written into what every agent reads damages every session at once. The question is therefore not whether deployed agents should improve themselves — at scale there is no alternative — but what discipline makes it safe enough to run in production: evidence before proposal, proof before release, staged exposure before full exposure, reversibility after, and an audit trail throughout.

SenseLab is an engine built on that discipline, and this paper measures it. Its loop has four levels (§2). At level 0, every session's outcome is attributed to exactly what the agent read and did, and that evidence — validated, contested, discredited — changes what the next session acts on. At level 1, an entry that was validated and now fails is a *regime shift*; a repair agent drafts a correction on a branch with a manifest of the runs it claims to repair and must not break, and the correction must pass a stand-in replay (Tier 1), the customer's own agent replaying the case set (Tier 2) and a live canary on a hashed share of sessions that must show the fix no worse than the rule it replaces (Tier 3) before it ships — reversibly. At level 2, the loop's own history (what it tried on this entry, what passed, what regressed) is evidence for its next draft. At level 3, the record the loop produces trains the next model and the repair prompt itself. The claim is not that the repair agent is clever. The claim is that **a correction may only change behaviour after it has been tested on the runs it claims to repair and shown no worse than the control arm on live traffic**, and that a system built this way recovers faster than one that can only demote what failed, while shipping fewer wrong corrections than one that ships on a passed test.

We ask, on a benchmark that knows the true rule before and after every change, whether the whole engine recovers faster and further than the frontier model alone, than three vector-store baselines, than a context window carrying the cell's entire history, and than its own ablations; whether it damages what still worked; what it ships; and what it costs. We call the whole system *SenseLab* throughout: the levels are not features to be compared against each other, they are what the engine delivers, and the ablations that switch parts of it off appear in one attribution table and nowhere else.

What this version is. This is the first version with the confirmatory read: seeds 37, 41, 59 and 73, both agent models, run once after the analysis code was tagged, with every endpoint and its test fixed beforehand (§5). The development-seed sections (§4) are kept as they were written, because the preregistration (§3) committed to reporting every prediction, met or falsified, and most of the predictions so far were falsified — by defects in the loop's evidence path that the benchmark found and that were fixed in production between runs. Section 6 is the record of that, and we think it is the paper's main contribution: a self-improving system is only as trustworthy as its evidence path, and every one of these defects would have been invisible to a benchmark that measured retrieval.

The two families of confirmatory cells stand differently: the fleet-1 cells are read and final; the ten-agent fleet-10 cells ran twice, because the first run exposed the last of the findings, and their second run is the read reported here.

2 The engine under test

Everything measured is production code on the SenseLab SaaS, run against a dedicated Enterprise account; no constant below was tuned for this study, and each is quoted from the deployed source. The harness never decides anything: it seals sessions through the public API, and the loop's own scheduler proposes, tests, gates and ships. The four levels of the loop, in the order evidence flows through them:

Level 0 — outcome learning: the evidence record. An agent calls `briefing` and `retrieve` before acting, `record_action` and `record_attempt` as it goes, and `commit_outcome` when the task ends. The sealed trace carries the exact read set; the outcome is attributed to it (an entry read by a run that failed after a recorded attempt against it is charged the failure; an entry read by a clean success is credited). Entries carry an *evidence status* — `untested`, `validated`, `contested`, `discredited` — and `retrieve` returns, with the hits, per-action priors (won/lost here on similar tasks) and a recommendation mode: `act`, `explore` (assigned per agent, so peers explore different actions) or `escalate`. That is level 0 alone; it is the row "SenseLab − repair loop" in Table 4.

Regime shift and situation. A *regime shift* is an entry validated at least three times that now fails: a sealed failure whose read set contains such an entry, or a recorded failed attempt against it. A shift is not a first-exposure failure — an entry that was never validated cannot have shifted — and the guard that separates the two was the third defect this study found (§6). Because the same entry can be right in one situation and wrong in another, the agent labels each session with an `attributes.situation`, and the service resolves the label to the agent's canonical situation (a stemmed-token rule, Jaccard ≥ 0.5, with an embedding second stage at cosine ≥ 0.75); the shift, the proof set and the canary are all scoped to the situation. A shift proposes a fix automatically (`auto_propose`, per agent).

Level 1 — the repair loop. Proposal. The repair agent (GPT-5-class, 50k-token budget) reads the failing runs, the entry, and the loop's own history on that entry, and writes a corrective entry (lever `knowledge`) or a conditional procedure with conditions, steps and a verify (lever `procedure`) on memory branch `repair/<fix>`, with a *manifest*: the runs it predicts it repairs, the runs it must not break, and an expected prevention (0.5 by default). A `code` lever is handed off and never auto-shipped.

Tier 1 — stand-in replay. A stand-in model re-runs each manifest case with the branch as memory and tool calls answered from the recorded run; fidelity is the share of steps answered from the record, and a test below the agent's fidelity floor (0.5) does not count as a pass. A test can be `passed`, `failed` or `inconclusive`.

Tier 2 — customer replay. The fix's *case set* — HMAC-signed cases — is delivered to the customer's agent, which re-runs them on the branch and seals the results; the service grades each replay against the case's expectation. For a regime fix the case set is the run the shift fired on, up to three more of the situation's failed runs (the shift; the fix must turn them into successes) and

up to three of its passed runs (what still works; the fix must not break them). A case no judge graded is graded on the outcome the replay reports, and — since the last cut — on its attempts: a success reached *after a recorded failed attempt* fails the case (§6, finding 9).

Tier 3 — canary. A passed fix opens a canary: sessions are bucketed by sha256(*fix*:*session*), a share `pct` reads from the branch, writes always go to `main`, and the trace is stamped with its branch. `decide()` requires both arms to reach `min_outcomes` graded runs — runs *in the fix's situation or that read an entry the fix is about* (cut *post-canary-ni*; before it, situation only: §6, finding 11) — and guards that the canary's success rate is not below the control's. The canary is the loop's *harm* gate: the replay tiers prove the benefit on the failing runs, and live traffic has to show the change does not make the agent worse on the runs the old rule still served. It passes when the canary arm recurs significantly less (one-sided two-proportion z-test, $p < 0.05$) or when the one-sided 95 % upper bound (Newcombe) on canary−control recurrence is under a margin of 10 points with the failure live in control; it fails when the canary arm recurs significantly more. Before *post-canary-ni* the primary asked for a significant improvement, and no canary ever passed (finding 11). The benchmark maps the production defaults (10 %, 20 outcomes, 14 days) to `pct`=50, `min_outcomes`=8, a timeout of 60 sessions (the cell horizon — 100 or 200 — from amendment (v) on), evaluated every 5 graded sessions. One canary per agent at a time.

Ship, verify, revert. Policy `auto_after_canary` merges a passed canary's branch into `main`; `auto_after_replay` (the "− canary" ablation) merges on a passed Tier 2. A feedback contract holds a fix if fewer than 80 % of recent traces carry provenance. Prevention is verified over the following window; a regressed fix is rolled back, restoring the pre-fix versions. Continuous judges (a two-tier `gpt-5-mini`→`gpt-5` rubric pack) grade sealed traffic and open incidents that are the second producer of proposals.

Level 2 — the loop reads its own history. Every fix event on an entry — proposed, tested, replayed, shipped, regressed, rolled back — is written back as evidence on that entry, and the repair agent's next draft on the same entry reads it: what was tried, what passed, what regressed. A loop that cannot see its own record repeats its own mistakes; this one is measured on whether it stops (the two-change protocol of the flagship experiment, not yet run).

Level 3 — the record trains the model and the improver. The traces the loop produces — including the repair trajectories and the cases it pinned — are a training set: Managed Models (`senselab-s1`) are distilled from them and served with the same evidence record as their memory (§4); and the repair prompt itself is A/B tested across proposals, with proven fixes per prompt as the score, which is the improver improving itself. Only the first rung of the first of these is measured in this version.

3 Benchmark and protocol

Preregistration. Hypotheses, arms, metrics, seeds, the dev/confirmatory split and the analysis freeze were registered before the first run (`HYPOTHESIS_RSI.md`, tag `prereg-rsi-v1`); every change since is a dated amendment (a)–(x) appended below the original, each with a prediction and what would falsify it. Seeds 11 and 23 are the development seeds; 37, 41, 59 and 73 run once, after the analysis code is tagged `analysis-freeze-rsi`. Engine changes made between runs are recorded with their deploy cut to the second, and every result carries the engine cut as a covariate.

Environment. Two families of the ten in the benchmark, both with tool-using agents and oracle-graded outcomes: `support` (a helpdesk agent with account, order and export tools; 15 issue classes) and `ci-fix` (an agent repairing failing CI on a repository; 10 failure classes). Every task instance is generated, never repeated, with random identifiers (`opaque_ids`) so nothing can be read off an id. 100 episodes per cell, 300 distractor entries in every store, retry budget 3, one agent identity per cell, GPT-5.5 at low effort, a post-episode reflection note in every arm that has a store.

Change set C. At episode 30 a subset of classes changes *conditionally*: the correct action moves for instances with one attribute value (the *moved half*) and stays for the others (the *kept half*). Half of post-change tasks are drawn from changed classes. Set C was written after the pilots showed that on an unconditional change (set A) outcome learning alone recovers nearly everything a maintainer who knew the rule would (§6, finding 4): the only place a written rule can beat demotion is where demotion has no signal to demote — the entry is still right half the time. Set C is registered by hash; its remaining eight families are written blind before the confirmatory run.

Systems. Frontier alone (no store); context window (every prior note and outcome of the cell in the prompt, most recent first, 24k tokens); pgvector (bge-small, cosine top-5 over notes); pgvector + rerank (the same cross-encoder SenseLab serves); pgvector + outcomes (notes, LLM consolidation every 5 episodes, a per-entry win/loss counter shown to the agent); and SenseLab (§2, policy `auto_after_canary`). Mem0 and Zep are not run (an earlier grid covers them [1]). The ablations — SenseLab − canary (`auto_after_replay`) and SenseLab − repair loop (policy `review`, proposals never approved) — appear in Table 4 only.

Endpoints. First-attempt success on the moved half after the change (primary); first-attempt success on the kept half and on unchanged classes (collateral damage); episode success and escalation; the fix lifecycle (proposed, Tier 1, Tier 2, canary, shipped, reverted); and cost. Infra errors are excluded from rates and reported. Intervals are cell-level bootstrap (2,000 resamples). With four cells per system on development seeds these intervals are wide by construction and are shown so that no reader mistakes a dev-seed difference for a result.

4 Results on the development seeds

4.1 After the rule changes conditionally

system	moved half	kept half	unchanged classes	pre-change	episode success	escalated	$/ep
Frontier alone	17 % (13/75) [0, 41]	38 % (25/65) [11, 78]	66 % (93/140)	61 % (73/120)	84 % (334/400)	16 % (66/400)	0.011
Context window	33 % (25/75) [0, 78]	98 % (64/65) [94, 100]	100 % (140/140)	85 % (102/120)	92 % (366/400)	8 % (34/400)	0.070
pgvector	17 % (13/75) [2, 38]	45 % (29/65) [13, 78]	93 % (130/140)	71 % (85/120)	88 % (353/400)	12 % (47/400)	0.027
pgvector + rerank	5 % (4/75) [1, 9]	86 % (56/65) [78, 95]	94 % (132/140)	79 % (95/120)	90 % (359/400)	10 % (41/400)	0.028
pgvector + outcomes (DIY)	19 % (14/75) [0, 44]	55 % (36/65) [39, 78]	80 % (112/140)	70 % (84/120)	88 % (351/400)	12 % (49/400)	0.035
SenseLab	33 % (25/75) [14, 56]	63 % (41/65) [40, 76]	96 % (134/140)	81 % (97/120)	92 % (368/400)	8 % (32/400)	0.033

Table 1: After the rule changes conditionally (set C, change at episode 30 of 100; GPT-5.5; four cells per system: `support` and `ci-fix` × seeds 11 / 23). First-attempt success on every post-change episode. *Moved half*: changed-class instances whose rule changed; *kept half*: changed-class instances whose rule did not. Brackets: cell-level bootstrap 95% interval. Development seeds; SenseLab on the `post-canary-ni` engine cut.

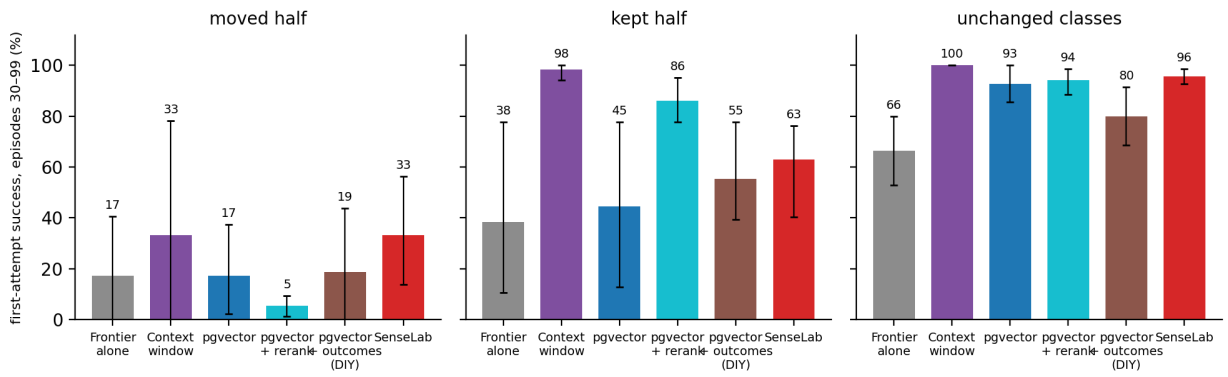


Figure 1: First-attempt success after the rule changes conditionally (episodes 30–99, four cells, GPT-5.5): the moved half, the kept half and the unchanged classes, one bar per system, with cell-level bootstrap 95 % intervals. Same numbers as Table 1.

Table 1 is the state of the engine on its current cut, `post-canary-ni` (§6, findings 8–11 live). Three things are in it. First, on the moved half every retrieval memory is at or below the frontier model alone: a note that says "do X for class C" is retrieved as confidently after the change as before, and pgvector with reranking — the best retriever — is the *worst* system on the moved half at 5 %, because it retrieves the stale note most reliably. Second, SenseLab is the only system that recovers the moved half on both families while holding the unchanged classes at 96 %: 33 % [14, 56] against 5–19 % for the frontier model and the three vector stores, winning every paired cell. The context window ties it exactly (33 %) by beating it on `ci-fix` (78 against 56: the whole history fits in 24k tokens and the model can see the change happen) and doing nothing on `support` (0/43, SenseLab 7/43), at 1.5× the all-in cost (2.1× the agent's own calls) and 3.7× the tokens. Third, SenseLab pays on the kept half: 63 % against 86 % and 98 % for the two systems that keep applying the old rule most reliably (pgvector + rerank and the context window). That cost is the shape of the mechanism — the entry is marked contested and the agent explores where the old action was still right — and it is the number the repair loop exists to remove: a written conditional rule would recover the moved half *without* giving up the kept one. Whether it does yet is Table 4.

pair	support s11	support s23	ci-fix s11	ci-fix s23	mean	cells won	p	p (Holm)
SenseLab − Frontier alone	+26	+5	+11	+23	+16	4/4	0.125	0.625
SenseLab − Context window	+26	+5	-26	-15	-3	2/4	0.875	0.875
SenseLab − pgvector	+22	+5	+11	+31	+17	4/4	0.125	0.625
SenseLab − pgvector + rerank	+22	+5	+47	+46	+30	4/4	0.125	0.625
SenseLab − pgvector + outcomes (DIY)	+26	+5	+5	+23	+15	4/4	0.125	0.625

Table 2: SenseLab minus each system on the moved half, per cell, in percentage points, paired on (family, seed).

system	support: moved	support: kept	support: unchanged	ci-fix: moved	ci-fix: kept	ci-fix: unchanged
Frontier alone	0 % (0/43)	78 % (21/27)	53 % (37/70)	41 % (13/32)	11 % (4/38)	80 % (56/70)
Context window	0 % (0/43)	96 % (26/27)	100 % (70/70)	78 % (25/32)	100 % (38/38)	100 % (70/70)
pgvector	2 % (1/43)	78 % (21/27)	86 % (60/70)	38 % (12/32)	21 % (8/38)	100 % (70/70)
pgvector + rerank	2 % (1/43)	78 % (21/27)	90 % (63/70)	9 % (3/32)	92 % (35/38)	99 % (69/70)
pgvector + outcomes (DIY)	0 % (0/43)	78 % (21/27)	69 % (48/70)	44 % (14/32)	39 % (15/38)	91 % (64/70)
SenseLab	16 % (7/43)	44 % (12/27)	96 % (67/70)	56 % (18/32)	76 % (29/38)	96 % (67/70)

Table 3: By family: first-attempt success on the moved half, the kept half and the unchanged classes.

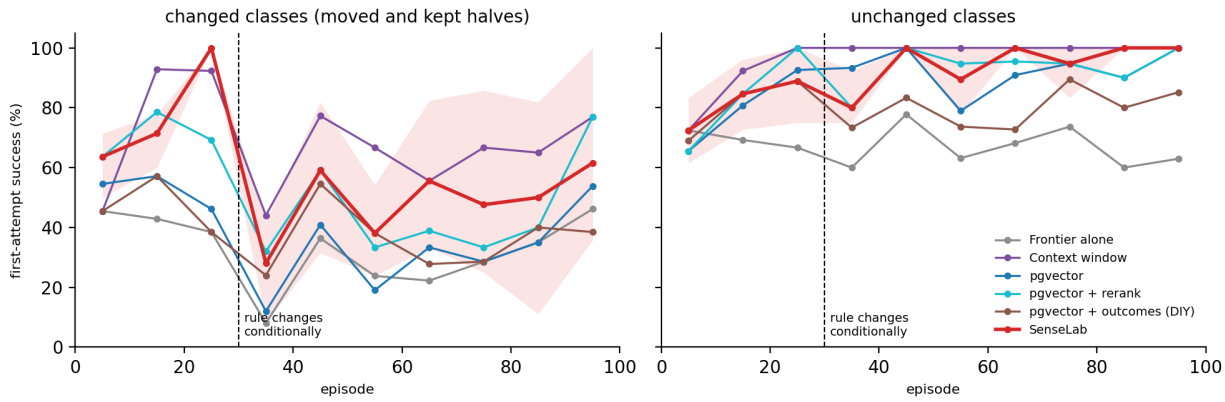


Figure 2: First-attempt success by 10-episode block across the change at episode 30, pooled over the four cells: the classes whose rule changes (left; before the change every system is at 80–90 %, after it the rule holds for the kept half only) and the unchanged classes (right). SenseLab is drawn heavy with its cell-level bootstrap band; the other systems' bands are of the same width and are omitted for legibility. Recovery is partial for every system inside 70 episodes; the context window recovers most on `ci-fix` because the whole history fits in its prompt (Table 3).

4.2 Attribution: what level 1 adds to level 0, so far

system	moved half	kept half	unchanged classes	pre-change	episode success	escalated	infra err	$/ep
SenseLab	33 % (25/75) [14, 56]	63 % (41/65) [40, 76]	96 % (134/140)	81 % (97/120)	92 % (368/400)	8 % (32/400)	0	0.033
SenseLab − canary	39 % (29/75) [14, 66]	62 % (40/65) [42, 74]	99 % (139/140)	79 % (95/120)	91 % (365/400)	9 % (35/400)	0	0.033
SenseLab − repair loop	51 % (38/75) [33, 73]	69 % (45/65) [58, 76]	96 % (135/140)	81 % (97/120)	92 % (369/400)	8 % (31/400)	0	0.032

Table 4: Attribution. SenseLab against its own ablations on the same protocol: without the canary (policy `auto_after_replay`, same run) and without the repair loop (policy `review`: proposals never approved; outcome learning alone). This is the only table in which the ablations appear.

On these seeds level 1 does not yet add to level 0. Without the repair loop (policy `review`: the loop proposes, nothing ships, the memory learns from outcomes alone) the moved half is 51 % and the kept half 69 %; with it, 33 % and 63 %; without the canary but with the replay gate, 39 % and 62 %. The intervals overlap throughout and nothing here is a difference. What the lifecycle table shows is *why*, and it is not that the loop is idle.

arm	proposed	regime / judge	T1 pass / inc. / fail	T2 pass / fail / open	canaries run / undecided	shipped
SenseLab, pre-seam4	9	5 / 4	7 / 2 / 0	2 / 2 / 0	3 / 3	0
SenseLab − canary, pre-seam4	8	4 / 4	4 / 3 / 1	1 / 2 / 0	0 / 0	1
SenseLab, post-seam5	16	12 / 4	5 / 10 / 1	6 / 3 / 7	4 / 3	0
SenseLab − canary, post-seam5	17	13 / 4	12 / 4 / 1	9 / 7 / 1	0 / 0	9
SenseLab, post-seam5b	24	19 / 5	14 / 9 / 1	15 / 8 / 1	6 / 4	0
SenseLab − canary, post-seam5b	16	13 / 3	10 / 5 / 1	11 / 4 / 0	0 / 0	11
SenseLab, post-clean-success	16	13 / 3	9 / 7 / 0	3 / 13 / 0	3 / 3	0
SenseLab − canary, post-clean-success	17	13 / 4	9 / 4 / 4	2 / 12 / 3	0 / 0	2
SenseLab, post-canary-ni	13	8 / 5	4 / 8 / 0	4 / 7 / 1	3 / 3	0
SenseLab − canary, post-canary-ni	14	10 / 4	4 / 9 / 1	0 / 13 / 0	0 / 0	0

Table 5: Fix lifecycle by run and arm. Tier 1: stand-in replay on the fix branch; Tier 2: the customer's replay of the case set; a canary grades live exposure in the fix's situation. *open*: Tier 2 replays still awaited when the cell ended. No canary reached a decision inside a 100-episode cell; no fix was reverted; 96–100% of proposals were drafted by the repair agent.

4.3 What level 1 did: the gating record

The repair loop did not add to the outcome endpoints on these seeds. What it did instead is the result we think matters for an engine that changes its own behaviour in production: it *refused*. Every fix it drafted went through the proof set and, if that passed, the live canary, and on the current cut nothing shipped — not because the loop was idle (it proposed 13 fixes in four cells) but because the proof rejected the drafts that restated or flipped the rule wholesale, and the canary held the rest because the failure they were written for was no longer live. The ablation that removes the canary shipped six wrong-conditioned rules on the fleet-10 run and none of them broke the kept half (§4.4); the ablation that removes the proof lets everything a stand-in replay cannot refute reach the canary. A self-improvement loop that ships only what it can prove and only against a failure it can still observe is slower than one that ships what looks right, and the record below is what that slowness bought and cost on the development seeds. It does not need more runs to state; the confirmatory seeds re-measure it on the same protocol (§5).

What the loop did in the cells. On the current cut the loop proposed 13 fixes in the four SenseLab cells, 8 of them from regime shifts; 4 passed Tier 1 (8 inconclusive), **4 passed Tier 2 and 7 failed it** (one still open), and none shipped: no canary decided before the cell ended. The three canaries that started were running at episode 100. Two, on regime fixes, had 2/8 and 4/8 canary-arm runs graded against the floor; the third, on a judge fix, was fully graded — 18 canary-arm and 31 control-arm runs — and still had no verdict, because at 6 % against 10 % recurrence the one-sided bound on the difference was 13 points against a margin of 10. That is the arithmetic of the harm gate at this sample: with a 10 % control recurrence the bound falls under the margin at about 15 graded runs per arm if the canary never recurs, about 30 if it recurs at 6 %, and about 50 if it recurs as often as control; a 100-episode cell supplied 18 and 31 to its best canary. A running canary is not inert: half of the sessions in its situation read the fix branch while it runs, so an unshipped fix reaches behaviour in SenseLab cells too. Two cuts earlier the same protocol passed 15 of 24 at Tier 2 and the ablation without the canary shipped 11 of 16; on the previous cut the ablation shipped 2 of 17; on this cut it shipped **none** of 14 (Tier 2: 0 passed, 13 failed), and its moved and kept halves are 66 % and 71 % on `ci-fix` (SenseLab: 56 and 76) and 19 % and 48 % on `support` (SenseLab: 16 and 44). With 27–43 episodes per half per family none of these is a difference, and the confirmatory run is powered for one where this one is not.

What the proof rejects. What the proof-passing fixes *say* is not in doubt, and what the proof now rejects is the result of findings 9 and 10. Three cuts earlier, hand-coding the fix text against the true change, 20 of the 24 regime fixes that passed their proof set encoded an unconditional flip ("always Y") where the change was conditional ("Y when a, else X"); on `ci-fix`, 19 of 19. The proof graded a replay on the *outcome* it reported — or on a judge that, by its frozen rubric, passes a recovered run — and this agent, like any agent with a retry budget, recovers: a fix that flipped the rule wholesale passed every case in which it could. Since the `post-clean-success` cut a proof case passes only on a clean success, whatever grades it. On that cut Tier 2 failed 19 of the 23 regime fixes that reached it (5 on a *kept* case, 10 on a *moved* case only, 4 on a pre-change or unchanged case); on the current cut, with the repair agent also handed the situation's passing runs (finding 11), it failed 15 of 17: 4 on a *kept* case (a wholesale flip broke the half the old rule still holds for), 10 on a *moved* case only (the old rule in better words did not repair the half that moved), 1 on a pre-change case. Both ways a correction can miss a conditional are caught. The two that passed — both in canary in SenseLab cells, neither shipped — are a wholesale flip on `ci-fix` ("prefer bumping … even if the changed files include related legacy code", the opposite of the true

condition) and, on `support`, a retry ladder (settings, then cache, then reinstall) whose evidence section names the ticket that needed the new action but whose rule branches on what has already failed, not on the app version the change turned on. Their moved and kept replays succeeded cleanly anyway: the replaying agent read the rule and did not follow it into the wrong action. A proof pass is therefore now necessary but cannot by itself certify a conditional when the agent's compliance with an entry is partial; that is what the live canary and the judge's followed-memory criterion are for, and the confirmatory analysis reads the criterion per episode to separate "the rule was wrong" from "the rule was wrong and followed".

What the drafts say now. Amendment (v) also changed the draft: the repair agent is shown the situation's passing runs and asked for the condition that separates them from the failing ones. P11(a′) asks whether at least half of the regime drafts whose proof set holds a kept case state a condition at all. Coded by the author against the written instructions (amendment (y): not two blind coders; every text, the coding and the key are in the reproducibility package), over the 37 such drafts of the two runs on this cut: **21 of 37 (57 %) state an input-dependent branch** — met. The split by class is the finding. On the CI family 11 of 14 drafts are conditional and every one of the 11 names the true variable (whether the PR touches `legacy_sdk/`): 4 state the rule exactly, 7 hedge it with a "try the bump first" step. On `support`, 9 of 23 are conditional, and the variable they branch on is the one visible in the ticket — platform (Android, iOS, web) or plan — not the one the world changed on: the plan is named by 2 of 7 export-timeout drafts, the issuer country by 0 of 8 card-declined drafts, the app version by 0 of 6 notification drafts (all six are retry ladders). So the drafter now writes conditions where the evidence makes the variable legible (file paths in a diff) and writes the salient attribute where it does not (a ticket), and the proof set is what separates the two: of the five drafts that passed Tier 2, the two that are conditional both name the true variable, and the three that are not are the flips and ladders above. Kept-case rejections, the second clause, were 4 of 17 on fleet 1 and 11 of 39 on fleet 10 — fewer than the 5 of 23 of the previous cut in count, not in share. The confirmatory fleet-1 runs (§5; 39 such drafts, the same sheet, instructions and key, coded the same way) read the same: **27 of 39 (69 %) state a condition** — met — and the class split holds. On `ci-fix` 20 of 28 are conditional and 17 of them branch on the PR's files, 13 naming `legacy_sdk/` or the cassette path exactly and 4 naming "an intentional legacy pin" without the path, or with the path and the actions inverted; the other 3 branch on whether the audit log names a fix version, or on PR intent. On `support` 7 of 11 are conditional and none names the true variable: they branch on platform (Android against web or iOS, 4 drafts), on whether a card is on file (2) or on whether the account is verified (1); the 4 unconditional drafts are the tier-1 playbook and three retry ladders. The drafts that pass on the CI family and fail on support are therefore the same drafts under the same instruction; the difference is where the evidence puts the variable in front of the drafter.

4.4 What the replay gate alone let through (exploratory)

No canary decided inside a cell, so the canary's value cannot be read from Table 4. What can be read, after the fact and outside the preregistration, is what happened to traffic once the no-canary ablation shipped: 21 ships over nine cells in three dev runs, every one a regime fix that had passed Tier 1 and Tier 2. Table 6 takes each such cell from its first ship to its end and compares the two halves of the changed class with the SenseLab cell of the same family, seed and cut over the same episodes — which shipped nothing.

population	−canary before	−canary after	SenseLab before	SenseLab after	DiD
kept	13/46 (28%)	29/99 (29%)	11/46 (24%)	43/99 (43%)	-18%
moved	50/59 (85%)	68/111 (61%)	46/59 (78%)	52/111 (47%)	+8%

Table 6: Exploratory, not preregistered: traffic after the no-canary ablation shipped, against the SenseLab cell of the same family, seed and cut over the same episodes (nine cells with a ship; window from the first ship to the end of the cell). *Recurrence* is the canary's reading of a session: not a first-attempt success. DiD = $(\text{after} - \text{before})_{-\text{canary}} - (\text{after} - \text{before})_{\text{SenseLab}}$; negative means the ships were associated with less recurrence than shipping nothing.

The ships did not break the kept half. Recurrence on the kept half stayed flat in the ablation (28 % to 29 %) while it rose in the SenseLab cell over the same episodes (24 % to 43 %), a difference of 18 points in the ablation's favour; on the moved half the ships were associated with eight points *less* recovery. Read with the hand-coding of §6 (20 of 24 proof-passing regime fixes restated the rule without its condition), this is one mechanism, not two: a correction that restates the old rule in better words protects the half for which the old rule still holds and slows the half for which it does not. It also names what level 0 does on its own when a validated entry starts failing conditionally: it demotes the entry for everyone, including the half it was still right for — the kept-half loss in Table 1 is that demotion, and the repair loop is the only level that can write the condition back. For the canary this cuts both ways. A tally over all traffic in the situation would have seen the kept half improve and the moved half worsen and found little to stop; the false-ship question the preregistration asks (H3) is whether the canary refuses corrections that damage the situation on net, and these did not. The canary's value has to be read where canaries decide — the fleet-10, 200-episode design of E2 — and against corrections that do net damage, which the adversarial probe supplies by construction. We report this analysis because it changed what we expect to find there.

Where canaries could decide. We then ran the E2 shape on development seeds (`rsi-e2b-c5`, cut `post-clean-success`): the same two families and seeds, ten agents sharing one store, 200 episodes with the change at 50, three arms (SenseLab, −canary, −replay), 2,400 episodes. On `ci-fix` every arm recovers both halves by episode 130 (moved half 93–100 %, kept 94–96 % over episodes 130–199; 70–77 and 83–90 over all post-change episodes); on `support` none does (moved 26–32 %, kept 51–63 %). The arms are indistinguishable at this sample. Twenty canaries started and none decided: thirteen were stopped by the 60-session timeout, three ended with no gradable run, four were running at episode 200. The reason is finding 11 (§6): of 16–30 exposed runs per arm, 0–8 could be graded, because grading is by canonical situation and the agent's declarations of its situation had been folded to other canonicals than the fix's. The −canary arm shipped four fixes (three regime); SenseLab and −replay shipped none. The canary's value therefore remained unmeasured, and the measurement waited on an engine correction rather than on a longer horizon.

The same run on the corrected cut. The rerun (`rsi-e2b-c6`, cut `post-canary-ni`; 2,400 episodes, 1 infrastructure error) is the fleet-10 re-measurement of finding 11 and the read of P12–P14 (§7). The outcome picture did not move: on `ci-fix` every arm recovers both halves by episode 130 (moved 86–93 %, kept 91–98 % over episodes 130–199; 71–77 and 81–92 over all post-change episodes); on `support` SenseLab's moved half is 38 % (31/82; 49 % over 130–199) against 35 and 22 for −canary and −replay, kept 59 against 65 and 74, and the three arms are indistinguishable at this sample. The −canary arm shipped six fixes (five regime); SenseLab and −replay shipped none. Nine canaries started, all on regime fixes, and Figure 3 is their time course from the sealed traces'

arm stamps. The correction did what it was for: the share of exposed runs the canary could grade rose from 0–27 % to 12–55 %, and four of the nine reached the floor of eight graded runs on both arms (P12's first clause: not met at 4 of 9, not falsified; its second, graded/exposed ≥ 0.8, is met by none). None decided (P13 falsified, 0 of 9). Of the four at the floor, three found **no recurrence in either arm** (26/11, 27/22 and 15/12 graded, 0 recurred) — by the time the canary could measure, the failure it was gating no longer occurred in control, and the rule that requires the failure live in control before a fix can ship (a fix for a failure that has stopped is not a fix) held it open. The fourth, on `support`, had 0 % against 10 % at 9 and 10 graded and a bound of 14 points against the margin of 10; it was also started at cell episode 8, forty episodes before the change, on a regime signal the change had not yet caused. Two ended with no gradable run and three were still short of the floor. So the bottleneck moved: from grading (finding 11) to recurrence. On these families and this horizon, once a fix reaches the canary the fleet has mostly stopped making the mistake — level 0's demotion and the agents' own recovery got there first — which is a statement about *when* the loop's evidence becomes available relative to when it is needed, and the reason the confirmatory fleet-10 cells carry the same read rather than a longer one. P14 is undefined (no verdict); read exploratorily, the kept-class first-attempt success in canary-arm against control-arm sessions on the two proof-passing `ci-fix` fixes with kept cases was 21 points *lower* on one (17/26 against 13/15) and 24 higher on the other (23/23 against 13/17): the sign the harm gate exists for, at a sample where it is noise.

What this read could not see. The paragraph above is kept as written on the day, because it is the reading the record supported until finding 12 (§6) was made the following morning from the loop's own read record: no canary-arm run in any of these nine canaries had read the fix's entry, and on the branch the arm had in fact *lost* the key. The recurrence the canaries compared was therefore that of two untreated arms, and "the fleet had stopped making the mistake before the gate could measure it" describes the agents' recovery without the correction, not the correction's timing. The bottleneck had not moved from grading to recurrence; it had moved from grading to a seam the canary sits behind. The confirmatory fleet-10 cells (§5) are the first run in which a canary arm holds the correction it is grading.

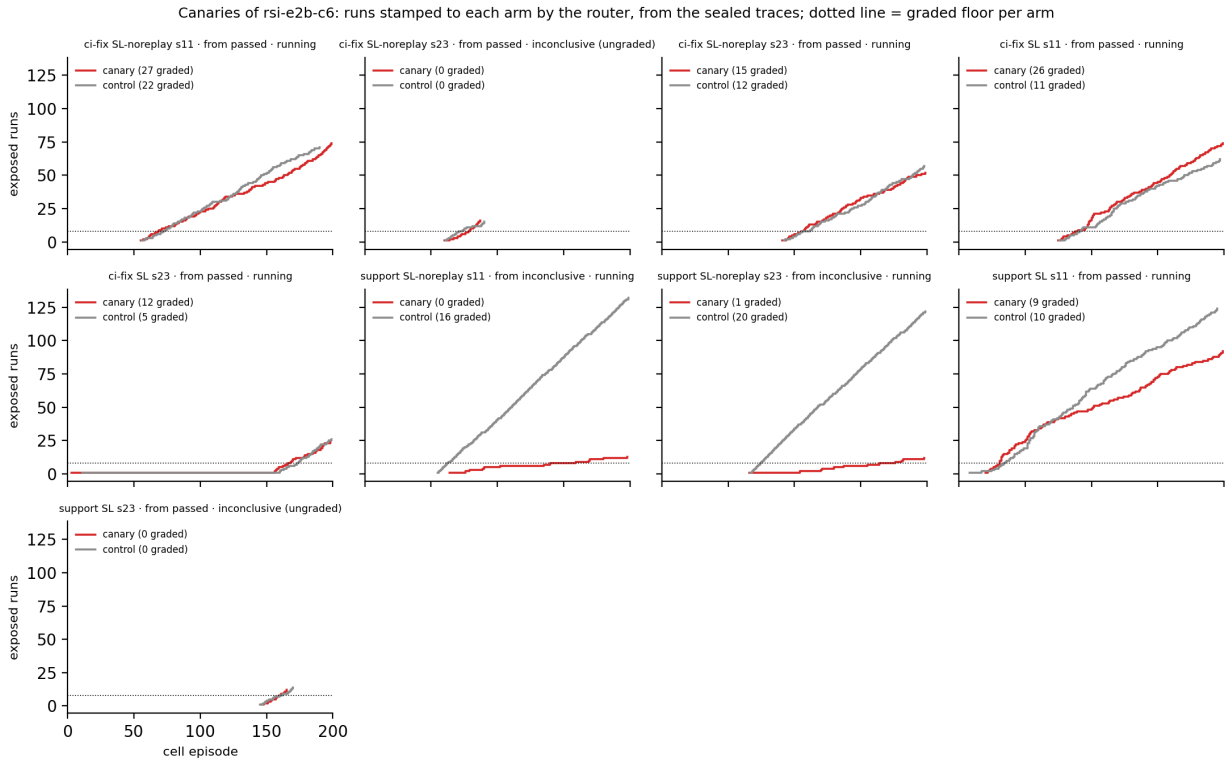


Figure 3: The nine canaries of `rsi-e2b-c6` (fleet 10, 200 episodes, change at 50), each panel one fix: runs the router stamped to the canary arm (red) and the control arm (grey) by cell episode, from the sealed traces; dotted line = the graded floor of 8 per arm; the legend gives how many stamped runs the canary could grade. Canaries started from an inconclusive Tier 1 route 10 % of traffic to the fix, the rest 50 %. Four reached the floor on both arms; none decided — three because the failure no longer recurred in control, one because 9 and 10 graded runs leave the non-inferiority bound at 14 points against a margin of 10.

4.5 SenseLab by engine cut

engine cut	moved half	kept half	unchanged	escalated	fixes proposed	Tier 2 passed	shipped
pre-seam4 (‘rsi-c1-rsi‘)	36 % (27/75)	65 % (42/65)	96 % (134/139)	6 % (22/399)	9	2	0
post-seam5 (‘rsi-c2‘)	47 % (35/75)	60 % (39/65)	99 % (138/140)	6 % (24/400)	16	6	0
post-seam5b (‘rsi-c3‘)	43 % (32/75)	60 % (39/65)	99 % (138/140)	8 % (30/400)	24	15	0
post-clean-success (‘rsi-c5‘)	34 % (25/74)	63 % (41/65)	96 % (132/137)	8 % (33/396)	16	3	0
post-canary-ni (‘rsi-c6‘)	33 % (25/75)	63 % (41/65)	96 % (134/140)	8 % (32/400)	13	4	0

Table 7: SenseLab on successive cuts of the engine, same protocol and seeds. Fix counts are over the four SenseLab cells of each run. `pre-seam4`: amfs-internal #787/#788; `post-seam5`: #793, #794; `post-seam5b`: #795, #796; `post-clean-success`: #797, #798; `post-canary-ni`: #799 (§6).

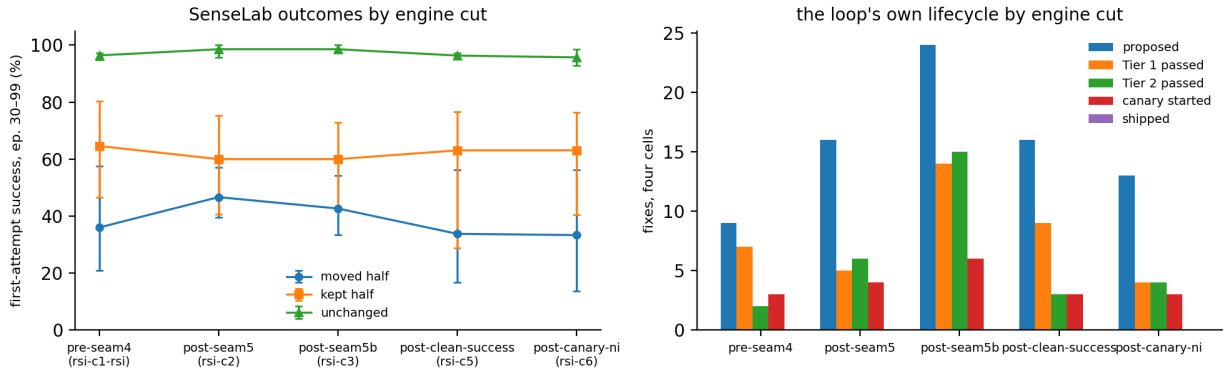


Figure 4: SenseLab on successive cuts of the engine, same protocol and seeds. Left: outcome endpoints, which move within their intervals. Right: the loop's own lifecycle counts over the four cells — what it proposed, what passed Tier 1 and Tier 2, how many canaries started and what shipped (nothing, on every cut: no canary decided inside a 100-episode cell).

Table 7 is the same protocol on successive cuts of the engine and is the trace of §6 in numbers: the loop went from 9 proposals and 2 Tier 2 passes on the first cut to 24 and 15 on the third as its evidence path was repaired, then to 16 and 3, and 13 and 4, as the proof learned to reject a recovered success — while the outcome endpoints moved within noise throughout. The cut under test now is the first on which the proof set can see the conditional and the drafter is shown the runs the old rule still serves.

4.6 Cost

system	tokens/ep	agent $/ep	arm $/ep	judges $/ep	replays $/ep	drafts/100 ep	drafts $/ep (est.)	total $/ep	memory s/ep	wall s/ep
Frontier alone	1,287	0.011	0.000	0.000	0.000	0.0	0.000	0.011	0.0	5
Context window	24,322	0.070	0.000	0.000	0.000	0.0	0.000	0.070	0.0	4
pgvector	4,475	0.027	0.000	0.000	0.000	0.0	0.000	0.027	0.0	9
pgvector + rerank	3,953	0.025	0.003	0.000	0.000	0.0	0.000	0.028	0.0	10
pgvector + outcomes (DIY)	5,406	0.029	0.005	0.000	0.000	0.0	0.000	0.035	0.0	9
SenseLab	6,629	0.033	0.000	0.004	0.004	3.0	0.004	0.046	14.8	25
SenseLab − canary	6,581	0.033	0.000	0.005	0.007	3.5	0.005	0.049	14.1	24
SenseLab − repair loop	6,537	0.032	0.000	0.000	0.000	3.8	0.005	0.037	10.5	20

Table 8: Cost and latency per episode, by where the call is charged. *agent $*: the agent's own calls; *arm $*: reflection or consolidation the arm ran per episode; *judges $* and *replays $*: the loop's judge verdicts and the Tier 2 replays the harness re-ran on the fix branch, exact from the run records; *drafts*: fixes the repair agent drafted per 100 episodes, priced at the account ledger's remainder per draft (repair agent, Tier 1 stand-in replay, lever classification, discovery — not persisted per fix, so an estimate; §4.6). *total* sums the five. *memory s*: time in memory calls against the production API.

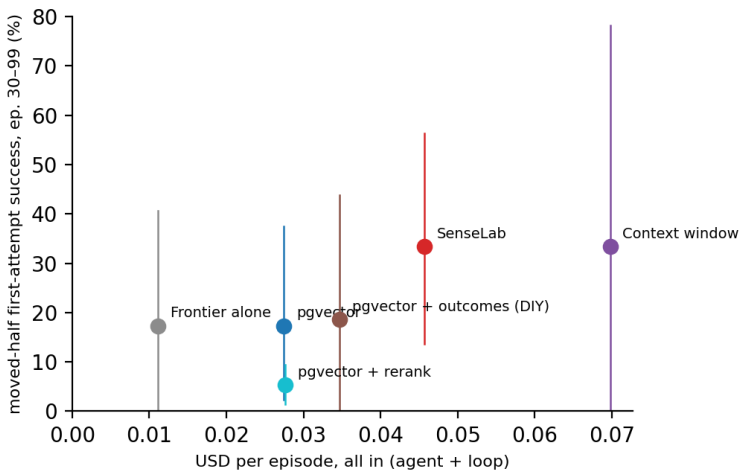


Figure 5: Recovery on the moved half against all-in cost per episode, the loop's own spend included (Table 8); vertical bars are the cell-level bootstrap intervals.

Every call is charged to the system that caused it, wherever it ran. The agent's own calls are in the episode records. The loop's platform-side calls are on the benchmark account's evaluation ledger, which settles every judge verdict, repair-agent step, stand-in replay, lever classification and discovery call as it happens; judge verdicts also carry their own cost, and every verdict and every Tier 2 replay of a cell is in the run record, so those two columns are exact per cell. The repair agent's spend is not persisted per fix. It is recovered as the ledger's remainder — what the account spent minus what its judges cost, over every fix the agent drafted on the account (USD 0.135 per draft at the time of writing, from USD 77.6 spent, 39.1 of it judges, over 285 drafts) — and applied per cell as drafts × that rate; the column says *est.* because it is one. Runs launched from the day the ledger was first snapshotted carry a reading at run start, after every cell and at run end, so the confirmatory runs report the ledger difference directly and this estimate is checked against it.

On these seeds SenseLab costs USD 0.046 per episode all in: 4.2× the frontier model alone (0.011), 1.3–1.7× the pgvector memories (0.027–0.035), and 0.66× the context window (0.070). A quarter of it is the loop (judges 0.004, Tier 2 replays 0.004, drafts 0.004 per episode); the rest is the agent reading a longer briefing than a top-5 retrieval. Holding the loop at review still drafts (3.8 proposals per 100 episodes, 0.037 per episode): the drafting is what a person would read. Memory time (15 s per episode against the production API, under the harness's rate pacing) is the price of reading the evidence record rather than a vector index.

4.7 Two probes on the earlier change set

When the team grows. On the unconditional change (set A, change at 20), with a fleet of six agents sharing one store: in the *pioneer* protocol one agent works alone for 60 episodes and five join; in the *newcomer* protocol five work and a sixth joins at 60. A joiner on SenseLab takes the post-change action on its first exposure to a changed class in 36/36 (pioneer) and 10/10 (newcomer) cases with zero stale picks; a joiner on pgvector + rerank does so in 31 % and 40 % with 17 % and 0 % stale picks and 16–20 % escalations. The inheritance is the outcome record, not the notes: the same notes are in both stores.

arm	veterans ep50-59 1st	veterans changed post 1st	joiner 1st-exp all	1st-exp changed	1st-exp unchanged	stale-pick (changed 1st-exp)	escalated
pgvector + rerank	0.75	0.28	0.68 [0.52,0.83] (n=90)	0.31 [0.15,0.50] (n=36)	0.93 [0.78,1.00] (n=54)	0.17	0.20
SenseLab	1.00	0.76	1.00 [1.00,1.00] (n=90)	1.00 [1.00,1.00] (n=36)	1.00 [1.00,1.00] (n=54)	0.00	0.00

Table 9: When the team grows (set A, change at 20, fleet of six sharing one store; *pioneer*: one agent works alone to episode 60, five join). First-attempt success of the joiners on their first exposure to each class after joining, with the veterans' rate before they joined; *stale-pick*: the joiner applied the pre-change action.

A student trained on the record (E7 at rung 0). Two Managed Models (`senselab-s1`) were trained on the dev-seed `support` traces of SenseLab and of the ablation without the repair loop (200 traces each), and evaluated on held-out seeds 101/113/127 (1,500 episodes). Alone, neither student takes the new action first (9 % and 0 % on the moved half); with SenseLab's outcome loop as its memory each does (33 % and 30 %), at 1/23 of the frontier model's cost per episode — and beats GPT-5.5 with the same memory (22 %) on the moved half at this rung, which is the contamination the evaluation ladder exists to expose: rung 0 uses the training classes. The SenseLab-trained student against the ablation-trained one is $-7/0/+18$ points across the three seeds — "small and reported as such" is the preregistered sentence. Nothing is claimed from rung 0; rungs 1–3, on classes and wordings the students never saw, need eval-only content that is registered but not yet written.

5 Confirmatory results

The seeds 37, 41, 59 and 73 run once, on both agent models, after the analysis code is tagged `analysis-freeze-rsi`; every table in this section is filled by that frozen code from the confirmatory read and nothing in it is known when the freeze is made. The endpoints, the paired test (exact two-sided sign-flip over the sixteen cells, Holm across the five systems on the primary endpoint, Holm within each secondary endpoint) and the tables are those of amendments (w) and (x); the development-seed tables of §4 are not re-read. The prose of this section is written after the read and states, for each preregistered prediction, whether it was met.

How the fleet-1 read was made. The four confirmatory seeds ran as four per-seed runs (`rsi-conf-f1-s{37,41,59,73}`, 3,200 episodes each, 01:42–08:19Z on 2026-09-30, all before the `post-canary-visible` cut of finding 12, so every fleet-1 canary arm is untreated by construction). The frozen reader resolves one run per system, so the four were concatenated file-by-file into `rsi-conf-f1` before the read — data staging, no code. Two wiring defects in the frozen reader, invisible on the development seeds because there the development runs *are* the resolved runs, were corrected before any confirmatory number was read from a complete table: the paired tables were wired to the development run names and came out empty under `--runs`, and the lifecycle table iterated the development cut list. Neither touches a metric, endpoint, test statistic or population rule; the development read was written to disk before and after and is byte-identical. The correction is commit `e04f78cf` on the paper branch, the one post-freeze change to analysis code, and the preregistration's read record names it. One engine cut fell inside the fleet-1 window: `post-canary-zero` (OSS #454, amfs-internal #800/#801, 03:40:19Z), which pins a fix branch's correction at retrieval and moves failed entries to the avoid list. 4,867 of the 12,791 episodes ran before it and 7,924 after; the moved-half difference between SenseLab and every other system has the same sign on the pre-cut episodes alone as on the full read (+28/+11/+6/+18/+23 pp against the five baselines, on 75 SenseLab episodes, against +19/+4/+13/+9/+13 on all), so the cut did not make the result; the split is in the package.

system	moved half	kept half	unchanged classes	pre-change	episode success	escalated	$/ep
Frontier alone	25 % (72/284) [13, 39]	27 % (75/276) [16, 39]	77 % (432/560)	65 % (312/480)	85 % (1362/1600)	15 % (238/1600)	0.014
Context window	40 % (115/284) [27, 54]	92 % (253/276) [84, 97]	97 % (542/560)	86 % (415/480)	92 % (1476/1600)	8 % (124/1600)	0.130
pgvector	32 % (91/284) [20, 45]	66 % (183/276) [59, 74]	96 % (539/560)	82 % (392/480)	92 % (1469/1600)	8 % (131/1600)	0.024
pgvector + rerank	36 % (101/284) [23, 48]	66 % (183/276) [58, 75]	91 % (511/560)	77 % (371/480)	90 % (1435/1600)	10 % (165/1600)	0.027
pgvector + outcomes (DIY)	32 % (90/284) [20, 44]	58 % (161/276) [48, 69]	84 % (470/560)	76 % (364/480)	88 % (1411/1600)	12 % (189/1600)	0.032
SenseLab	45 % (127/283) [39, 51]	69 % (191/276) [62, 76]	100 % (556/557)	85 % (406/480)	95 % (1516/1596)	5 % (80/1596)	0.033

Table 10: Confirmatory: after the rule changes conditionally. Sixteen cells per system (`support`, `ci-fix` × four seeds × two models); first-attempt success on every post-change episode; brackets: cell-level bootstrap 95% interval. Confirmatory seeds 37 / 41 / 59 / 73, both agent models (GPT-5.5, Claude Opus 5), the two set-C families, run once on the `post-canary-ni` cut after `analysis-freeze-rsi` (amendment (w)).

pair	support gpt-5.5	support claude-opus-5	ci-fix gpt-5.5	ci-fix claude-opus-5	mean	cells won	p	p (Holm)
SenseLab − Frontier alone	+28	+56	-9	-1	+19	11/16	0.024	0.121
SenseLab − Context window	+17	+36	-17	-19	+4	8/16	0.544	0.544
SenseLab − pgvector	+23	+16	+22	-11	+13	11/16	0.040	0.159
SenseLab − pgvector + rerank	+24	+7	+3	+4	+9	9/16	0.119	0.239
SenseLab − pgvector + outcomes (DIY)	+23	+34	-6	+1	+13	10/16	0.040	0.159

Table 11: Confirmatory primary endpoint: SenseLab minus each system on the moved half, percentage points, paired on (family, seed, model); per family × model means, the mean over the sixteen cells, cells won, the exact two-sided sign-flip p over the cells and its Holm adjustment across the five systems (amendment (x)).

Primary endpoint: the moved half. On the half of the changed classes whose rule moved, SenseLab's first-attempt success after the change is 45 % (127/283, cell-level interval [39, 51]) against 25–40 % for the five baselines. Paired on (family, seed, model) over the sixteen cells, the mean delta is positive against every baseline — +19 pp over Frontier alone, +4 over the context window, +13 over pgvector, +9 over pgvector with a reranker, +13 over the DIY outcome store — and SenseLab wins 8 to 11 of the 16 cells in each pairing. The exact sign-flip p is 0.024 against Frontier alone and 0.040 against pgvector and the DIY store, 0.119 against the reranker and 0.544 against the context window; *none survives Holm across the five systems at 0.05* (0.121, 0.544, 0.159, 0.239, 0.159). The preregistered primary prediction — SenseLab ahead of every baseline on the moved half — is met in sign and in cells won, and not at the preregistered significance. Sixteen cells is the power the reduced confirmatory scope of amendment (w) bought; the deltas are of the size the development seeds showed, and the reader should weigh them as estimates with the intervals above rather than as a rejection.

pair	support gpt-5.5	support claude-opus-5	ci-fix gpt-5.5	ci-fix claude-opus-5	mean	cells won	p	p (Holm)
SenseLab − Frontier alone	+1	+23	+53	+75	+38	14/16	0.002	0.007
SenseLab − Context window	-36	-12	-26	-21	-24	1/16	0.000	0.001
SenseLab − pgvector	+2	+8	-13	+9	+1	9/16	0.834	1.000
SenseLab − pgvector + rerank	-2	+1	+1	+3	+1	9/16	0.882	1.000
SenseLab − pgvector + outcomes (DIY)	-17	+3	+23	+25	+8	10/16	0.236	0.708

Table 12: Confirmatory secondary endpoint, the kept half: SenseLab minus each system, paired as in Table 11; Holm within this endpoint.

pair	support gpt-5.5	support claude-opus-5	ci-fix gpt-5.5	ci-fix claude-opus-5	mean	cells won	p	p (Holm)
SenseLab − Frontier alone	+37	+35	+19	+0	+23	12/16	0.000	0.002
SenseLab − Context window	+4	+9	-1	+0	+3	4/16	0.125	0.125
SenseLab − pgvector	+4	+4	+0	+6	+4	6/16	0.031	0.062
SenseLab − pgvector + rerank	+21	+14	-1	+0	+9	8/16	0.012	0.035
SenseLab − pgvector + outcomes (DIY)	+26	+25	+12	+0	+16	11/16	0.001	0.004

Table 13: Confirmatory secondary endpoint, the unchanged classes: SenseLab minus each system, paired as in Table 11; Holm within this endpoint.

Secondary endpoints: collateral damage. On the kept half — changed classes whose rule did *not* move — SenseLab is at 69 % (191/276), above Frontier alone by 38 pp (14/16 cells, Holm 0.007), level with the three pgvector arms (deltas +1, +1, +8; none distinguishable), and *below the context window by 24 pp* (1/16 cells, Holm 0.001). The context window carries every prior note and outcome of the cell in the prompt; on a rule that did not change, replaying the whole record is the right policy and it wins, at $0.130 per episode against SenseLab's $0.042. This is the cost of the moved-half gain on this protocol and it is reported as such: SenseLab's memory is selective and outcome-weighted, and on the kept half selection loses to exhaustive recall. On the unchanged classes SenseLab is at 100 % (556/557), above Frontier alone (+23, Holm 0.002), the reranker (+9, 0.035) and the DIY store (+16, 0.004), and not distinguishable from the context window or plain pgvector; no baseline is above it.

system	support: moved	support: kept	support: unchanged	ci-fix: moved	ci-fix: kept	ci-fix: unchanged
Frontier alone	0 % (0/150)	52 % (68/130)	64 % (179/280)	54 % (72/134)	5 % (7/146)	90 % (253/280)
Context window	17 % (25/150)	92 % (119/130)	94 % (262/280)	67 % (90/134)	92 % (134/146)	100 % (280/280)
pgvector	22 % (33/150)	62 % (80/130)	96 % (269/280)	43 % (58/134)	71 % (103/146)	96 % (270/280)
pgvector + rerank	27 % (40/150)	66 % (86/130)	82 % (231/280)	46 % (61/134)	66 % (97/146)	100 % (280/280)
pgvector + outcomes (DIY)	14 % (21/150)	74 % (96/130)	74 % (208/280)	51 % (69/134)	45 % (65/146)	94 % (262/280)
SenseLab	42 % (62/149)	69 % (90/130)	100 % (277/277)	49 % (65/134)	69 % (101/146)	100 % (279/280)

Table 14: Confirmatory, by family: first-attempt success on the moved half, the kept half and the unchanged classes.

system	gpt-5.5: moved	gpt-5.5: kept	gpt-5.5: unchanged	claude-opus-5: moved	claude-opus-5: kept	claude-opus-5: unchanged
Frontier alone	25 % (36/142)	28 % (39/138)	72 % (201/280)	25 % (36/142)	26 % (36/138)	82 % (231/280)
Context window	36 % (51/142)	88 % (122/138)	98 % (275/280)	45 % (64/142)	95 % (131/138)	95 % (267/280)
pgvector	13 % (18/142)	63 % (87/138)	98 % (274/280)	51 % (73/142)	70 % (96/138)	95 % (265/280)
pgvector + rerank	22 % (31/142)	57 % (79/138)	90 % (251/280)	49 % (70/142)	75 % (104/138)	93 % (260/280)
pgvector + outcomes (DIY)	27 % (38/142)	54 % (74/138)	80 % (225/280)	37 % (52/142)	63 % (87/138)	88 % (245/280)
SenseLab	36 % (51/142)	60 % (83/138)	100 % (279/280)	54 % (76/141)	78 % (108/138)	100 % (277/277)

Table 15: Confirmatory, by agent model: first-attempt success on the moved half, the kept half and the unchanged classes.

By family and by model. The moved-half gain is a `support` result. There SenseLab is at 42 % (62/149) against 0–27 % for every baseline — Frontier alone at 0/150 — while on `ci-fix` it is at 49 % (65/134) against 43–67 %, behind the context window (67 %). By model, SenseLab is level with the context window on GPT-5.5 (36 % both) and ahead on Claude Opus 5 (54 % against 45 %, the next best); the pgvector arms swap order across the two models (13–27 % on GPT-5.5, 37–51 % on Opus). Every system's kept-half and unchanged-class rates are within a few points across models; the model effect is on the moved half.

system	moved half	kept half	unchanged classes	pre-change	episode success	escalated	infra err	$/ep
SenseLab	45 % (127/283) [39, 51]	69 % (191/276) [62, 76]	100 % (556/557)	85 % (406/480)	95 % (1516/1596)	5 % (80/1596)	4	0.033
SenseLab − canary	44 % (125/283) [36, 54]	71 % (197/276) [66, 76]	99 % (548/556)	86 % (412/480)	94 % (1506/1595)	6 % (89/1595)	5	0.034
SenseLab − repair loop	43 % (122/284) [33, 52]	68 % (187/276) [59, 76]	99 % (557/560)	85 % (410/480)	93 % (1494/1600)	7 % (106/1600)	0	0.033

Table 16: Confirmatory attribution: SenseLab against its own ablations on the same sixteen cells. The only confirmatory table in which the ablations appear.

arm	proposed	regime / judge	T1 pass / inc. / fail	T2 pass / fail / open	canaries run / undecided	shipped
SenseLab, confirmatory	32	23 / 9	16 / 13 / 3	6 / 22 / 2	5 / 3	2
SenseLab − canary, confirmatory	44	32 / 12	25 / 14 / 4	10 / 32 / 0	0 / 0	10

Table 17: Confirmatory fix lifecycle by arm over the sixteen fleet-1 cells (the fleet-10 cells are read in §5).

Attribution and the loop's lifecycle. Inside a 100-episode cell at fleet size 1, the two ablations are indistinguishable from the full system: without the canary 44 / 71 / 99 % (moved / kept / unchanged), without the repair loop 43 / 68 / 99, against 45 / 69 / 100 — the same reading as on the development seeds, and the reason the long-horizon protocol below exists. The loop was active: 32 fixes proposed in the sixteen SenseLab cells (23 from a regime shift, 9 from the judges), 16 passing the stand-in replay, 6 the customer replay, 5 canaries opened, 3 still undecided when their cell ended, 2 shipped, 0 reverted. With the canary removed the same loop shipped 10 of 44 — the canary gate holds back four fixes in five that the replay would have shipped, in cells where none of them could have been graded to a decision.

The long horizon: fleet 10, 200 episodes. The E2b protocol on the four confirmatory seeds runs SenseLab alone on GPT-5.5 (amendment (y): 400 episodes per seed; the two gate ablations and the Opus cells are not run, since two development runs already read them the same way — arms indistinguishable, no canary decides). It supplies the learning curve on held-out seeds, descriptively, and the canary readout of amendment (v) on whatever canaries SenseLab starts: P12 (canaries reach the graded floor on both arms), P13 (canaries decide), P14 (no pass harms the kept class by more than the margin, joined to the sealed traces' arm stamps), P13′/P13″/P15′/P16 of amendments (z) and (z′) (the three classes; canary-arm reads of the fix), and P11(a′) (drafts state their condition; author-coded as in §4.3). The protocol ran twice on the confirmatory seeds. The first four cells (`rsi-conf-e2b-s{37,41,59,73}`, 400/400 each, 08:26–11:21Z on the `post-canary-visible` cut) are the read of finding 13 (§6): the arm read its fix for the first time in the project, and the loop's own replays discredited five of the six corrections within minutes, two before their canary's first session. They are reported there as measured and are not the confirmatory fleet-10 read. The confirmatory read is the rerun on the `post-replay-evidence` cut (amendment (aa); `rsi-conf-e2b-r2-s{37,41,59,73}`, launched 12:59–13:00Z on 2026-09-30, the same protocol and the same predictions; 400/400 episodes each, finished 15:45–16:08Z).

The confirmatory fleet-10 read. Over the eight cells the loop proposed 48 fixes (39 from a regime shift, 9 from the judges; 44 drafted by the repair agent, 4 from a template); the stand-in replay passed 23 and was inconclusive on 23; the customer's replay passed 6 and failed 38; 6 canaries opened, in six different cells; 0 fixes shipped, 0 were reverted. **The canary arm now reads its fix.** In the sessions the canary graded, the fix's own key was a version-exact read in 57 of 76 (75 %; per canary 50, 100, 75, 100, 47 and 100 %), and the control arm read it in 0 of 54 — P16 met (threshold 50 %; the engine's per-arm `read` count and the version-exact join agree within three sessions on every canary). Reads per canary arm at close were 2, 5, 8, 12, 14 and 18, median 10

against the floor of 8 — P13″ met, four of the six arms at or above the floor (finding 12's run had 0, 0, 1, 1, 1 and 17). **P12** is met: 4 of 6 canaries reached the graded floor on both arms (10/10, 20/8, 14/10 and 17/14 graded sessions); the graded share of the sessions routed to the arm was 0.23–0.36 on five of them and 1.0 on the sixth, so the second clause of P12 (≥ 0.8) is not met, for the structural reason finding 12 gives: exposure counts the arm's routing, grading counts the situation. **P13′** is not met and not falsified: one canary of six decided within the cell — `8a908156` (`support`, seed 37, "Android free-plan push notifications stopped"), *failed* at 10/10 graded sessions on the zero-count bound (control had no recurrence; the canary arm had), with the fix read in 5 of the arm's 10 graded sessions, below the read floor, so by the class rule of amendment (z″) it is a verdict on a partly treated arm and does not bear on P13′; the falsifier (0 verdicts in ≥ 6 canaries) is not met by the letter and is by the class rule, and we report both. The other five were open when their cells ended: three with both arms above the floor (20/8, 14/10, 17/14) and no decision — at these counts the non-inferiority test of finding 11 cannot close a 10-point margin either way, which is the gate's power at this horizon, not a defect — and two with the control arm or the canary arm below it. On the treated population, kept-class first-attempt success in canary-arm sessions that read the fix, against control sessions in the same situation, was 1.00 (6) against 0.88 (17) and 1.00 (6) against 0.77 (22) on the two `ci-fix` canaries that closed undecided with a treated population, 0.33 (6) against 0.40 (25) and 0.00 (3) against 0.70 (10) on the two `support/ci-fix` canaries at seed 41, all under the 8-session floor of P15′ (underpowered, reported as such); P14 and P15 have no population, since nothing passed. Live evidence reached the corrections this time. Five of the six fixes wrote a correction (the sixth, the one that decided, wrote corrective entries only), and every one of the five was charged by canary-arm sessions that had read it: at close 14 wins and 0 failures (0.82), 12 and 2 (0.65), 8 and 2 (0.47 — discredited at the second failure and pinned all the same), and, for the two rewrites of the agent's own entries, born at 0.37 and 0.30 under the 0.8× rule, 7 and 26 and 1 and 3, both discredited within two minutes of their first read. A correction born under the discredit threshold is discredited by its first failed attempt; the pin now keeps it as the arm's treatment regardless, and whether that threshold should apply to a correction under test is an engine question this version leaves open. No replay wrote an outcome row or a contrast lesson after the cut: 209 replays sealed in the window, 0 of 1,600 outcome rows from them. **The long horizon** (descriptive, per amendment (x); pooled over the eight cells with a cell-level bootstrap): moved-half first-attempt success 37 % (40/109) in episodes 50–99, 55 % (60/110) in 100–149 and 75 % (67/89) in 150–199 — 88 % (15/17) in the last block; kept half 66, 76 and 72 %; unchanged classes 100 % in every post-change block (594/595). The first run of the same cells, on the cut before, read 32, 59 and 67 % on the moved half. At 200 episodes with ten agents the system that could not be separated from its level-0 ablation inside a 100-episode cell reaches three quarters of the moved half, and the canary — now treating the arm it grades — holds every one of the six fixes it opened: five for want of power, one on a failure it saw.

system	tokens/ep	agent $/ep	arm $/ep	judges $/ep	replays $/ep	drafts/100 ep	drafts $/ep (est.)	total $/ep	memory s/ep	wall s/ep
Frontier alone	2,389	0.014	0.000	0.000	0.000	0.0	0.000	0.014	0.0	6
Context window	29,871	0.130	0.000	0.000	0.000	0.0	0.000	0.130	0.0	6
pgvector	5,005	0.024	0.000	0.000	0.000	0.0	0.000	0.024	0.0	10
pgvector + rerank	5,164	0.025	0.002	0.000	0.000	0.0	0.000	0.027	0.0	13
pgvector + outcomes (DIY)	5,784	0.027	0.005	0.000	0.000	0.0	0.000	0.032	0.0	11
SenseLab	7,856	0.033	0.000	0.003	0.003	1.9	0.003	0.042	18.4	31
SenseLab − canary	8,001	0.034	0.000	0.004	0.005	2.4	0.003	0.045	18.0	31
SenseLab − repair loop	7,983	0.033	0.000	0.000	0.000	0.0	0.000	0.033	16.8	30

Table 18: Confirmatory cost and latency per episode, columns as in Table 8; the drafts column is checked against the ledger difference the runs recorded.

Cost.

6 What the loop taught us about itself

The preregistration promised to report every prediction. Table 19 is that report, and the findings below are the record of what a benchmark that knows the true rule found in a shipped self-improvement loop — eleven defects and two properties of the problem — none of which a retrieval benchmark could have seen. All eleven were fixed in production between runs, with their deploy cuts recorded, and re-measured on the same protocol; the last three were found in the fleet-10 runs, the last two by the loop's own read record while the confirmatory fleet-10 cells were running, and the confirmatory fleet-10 read (§5) is their re-measurement. We think this section is the paper's main contribution; the loop's evidence path is where such systems fail, and none of these failures is specific to ours.

1. **The loop was never reached.** (Pilots 5b–5g.) Incidents open at three failing verdicts on one behaviour cluster; a fresh agent's judges had graded nothing, so no incident opened and no fix was proposed in 60 episodes. The regime-shift producer — an entry validated ≥ 3 times that now fails — is the producer a memory can drive without judges, and it is what every result here rests on.

2. **Sealing stalled under load.** A lock on the seal path held sessions open under the grid's concurrency; fixed (#780/#781) before any reported run.

3. **A first-exposure failure is not a regime shift.** (Amendment (k), pilot 5k; P1 falsified 3/3.) The shift producer fired on entries that had been *read*, not validated, so the first failure of a never-tested entry opened a canary that could grade nothing — the canary compares recurrence of a failure the control arm had never had. The guard now requires validation in the situation.

4. **On an unconditional change, demotion recovers almost everything.** (Amendment (l); P4.) A scripted maintainer writing the true rule five episodes after the change beat outcome learning alone by 2.5 (`support`) and 4.2 (`ci-fix`) points on 60 episodes of set A. That is the ceiling a repair loop can reach on such a change, and it is small. Change set C was written so that demotion has no signal: the entry is still right on the kept half.

5. **A regime fix had nothing to be proved against.** (Seam 4, #793; P6 met on the ablation.) Case sets were built from judge verdicts; a fix drafted from a shift had none, so Tier 2 could not be asked and `auto_after_replay` had nothing to ship on. The manifest's runs now seed the case set, and a judge-less case is graded on the replay's outcome.

6. **The agent spells one situation ten ways.** (Seam 5, #794; P7 recall 0.71, purity 0.92 against 0.80/0.90 predicted; seam 5b, #796, P7′ falsified: 0.40/0.67.) A canary scoped to a situation graded 2 of 15 in-situation runs because the agent labelled the same situation `csv export timeout`, `data export error`, `large export failing`. A server-side canonical resolves labels by stemmed tokens; the embedding second stage added for recall *lowered* it and merged `ci-fix` classes (`format check failure` $\sim$ `audit check failure` at 0.79). It is left on by decision, with its cost stated.

7. **A running canary swallowed every later fix's replays.** (Seam 7, #795; P8 met.) The router bucketed a Tier 2 replay session like any other; a replay for fix B sealed during A's

canary was stamped onto A's branch, counted in A's arm, and reported `missing` to B. Replays are now re-stamped onto their own branch and excluded from every canary tally.

8. **The loop drafted fixes from its own tests.** (Seam 8, #797; P10 pending.) A Tier 2 replay that failed on a fix branch read to the regime pass as the entry failing on `main`: 13 of 40 fixes on the `post-seam5b` cut were drafted from a replay, in chains — a fix from the replay of a fix from a replay — and one replay's spelling became a canonical that four served runs then joined. Replays are no longer traffic at the seal: no judge, no identity, no shift. (The first cut of this rule also excluded the labelled copy of a served inference, which is traffic; an automated reviewer caught it before merge.)

9. **The proof graded the wrong signal.** (Amendment (s), #797; P9(b) falsified 20/24, P11 pending.) See §4: an outcome-graded case passed a recovered success, so an unconditional flip passed a proof set built to catch it. A case now passes only on a clean success — no recorded failed attempt — and the change is dated by the case, not the deploy, so earlier case sets keep their semantics.

10. **Who grades the proof.** (Found while this version was being written; fix pending approval.) The clean-success rule of finding 9 applies to a case no judge graded. In this benchmark 174 of 175 proof cases carry a judge — the benchmark's own frozen rubric, which the continuous-judging tier had already run on the evidence trace — and that rubric says, by design, that "a run that failed one or more attempts before succeeding is a PASS". The proof tier therefore inherits the coarse signal through the judge as well as through the outcome, and finding 9's fix reaches one case in 175 here. The size of the leak is measurable: of the 555 case replays run since the `post-seam5b` cut, 479 passed, and 116 of those passes (24 %) carry a recorded failed attempt — a recovered success the proof counted as clean. The engine-side correction is to carry the clean-success requirement on every case built for a regime fix, whatever grades it, and enforce it after the verdict. It shipped as the `post-clean-success` cut; P11 is read on the run on that cut (§7), and Table 7 carries it as a row.

11. **The canary grades by canonical situation, and the agent's declarations fragment.** (Found in the fleet-10, 200-episode run of §4.4; fixed in #799 and re-measured on both protocols.) A canary counts exposure — sessions the router stamped to its arm — but grades only runs whose canonical situation equals the fix's. The agent declares its situation per run in its own words, often carrying the very condition the change turns on ("pip audit requests pin with legacy sdk", "pip audit requirements plus nonclient code"); situation identity folds one agent's 56 declarations to 10 canonicals, the pip-audit ones to "pip audit failure" — while the fixes carry "pip audit pinned change" and two more, assigned when each was proposed. So 0–8 of 16–30 exposed runs per arm are graded, three canaries end "none declared it", and no canary decides in 200 episodes with ten agents; the one whose declarations were stable reached 25/30 graded, found recurrence of 12 % against 17 % — no significant difference — and timed out, because the rule waits for a significant *improvement*. This, not the horizon, is why no canary has decided in any run. Corrected after approval (amendment (v), PR #799, cut *post-canary-ni*): the canary grades runs in the fix's situation *or* that read an entry the fix is about, and its primary is non-inferiority — a one-sided 95 % Newcombe bound on canary−control recurrence under 10 points, with the failure live in control — because the canary is the harm gate on a change the replay tiers already proved. Canonical merging was not done: it changes situation identity for every customer with the opposite failure mode, and the canary no longer depends on it. Pre-registered on the rerun: P12 (canaries reach the floor), P13 (canaries decide), P14 (no

pass harms the kept class) and P11(a′) (drafts state their condition; the regime draft now sees the situation's passing runs). Re-measured (§4.4): graded runs rose from 0–27 % of exposure to 12–55 %, four of nine canaries reached the floor, and none decided — three because the failure had stopped recurring in control by the time the canary could measure it. The grading defect is closed. What we read into the rerun at the time — that the canary's evidence arrives after the fleet has recovered on its own — was a reading of two untreated arms, and finding 12 supersedes it.

12. **The fix was written where no agent could read it.** (Found 2026-09-30 in the loop's own read record, after the confirmatory fleet-10 cells had run half their length with canary-arm reads of the fix still at zero; fixed in amfs-internal #802, cut `post-canary-visible`; the confirmatory fleet-10 cells re-run on it.) Every write the repair loop makes — the correction on the fix's branch, the corrective entry, the merged version once a fix ships, and the contrast lessons the outcome loop writes at every `commit_outcome` — is authored by the platform itself, as `amfs-server`. The per-user visibility rule that keeps one customer's agents from reading another's admitted an entry only when its author was one of the caller's own agents, or a room member or the platform on a room's shared path. The platform's own writes on an agent's ordinary path matched neither clause, so for every user-scoped API key — which is every key the hosted service issues to an agent — they did not exist. The engine's instrumentation is what made this visible: the reuse record stores, for every retrieve, read and search, which entry *version* the session was served, and joining it version-exactly to the arm stamps on the sealed traces gave the count. Of 25,875 reuse events over the two days before the fix, none was on a platform-authored entry; of every causal entry and read in every canary arm since the `rsi-e2b-c6` cut, none was the fix's; the 230 contrast lessons on `main` had never been read either. (A first count of 14 such reads joined to the entry's *latest* version and was wrong; the version-exact join gives zero.) On a canary branch the effect was worse than absence: the corrected key's branch version, being the platform's, replaced the agent's version of the key and was then dropped, so what the canary arm received was the key's *deletion*. Every canary in §4.4 therefore compared an arm that had lost the entry with an arm that still had the old one; the pin of #454 and the read gate of #801 had nothing to act on, and the three canaries that found no recurrence in either arm were reading two arms that had received no correction. The rule now admits a platform-authored entry on a path where, in the entry's namespace, one of the caller's own agents has written — never account-wide, so a member who owns no agents still reads nothing beyond their rooms — which on the benchmark account covers 318 of 318 corrections and 168 of 168 corrective entries. One limitation survives the fix and is reported rather than changed: the loop writes a correction at 0.8× the confidence of the version it replaces (floor 0.3) while the benchmark's agents read at a confidence gate of 0.5, so of the 31 shipped fixes that carry a correction, 20 sit below the gate on `main` — visible now, but not read there — while every corrective entry (0.7 or 0.75) is above it; on the canary branch the pin carries the correction to the head regardless. The re-measurement on the corrected cut (the four `rsi-conf-e2b` cells, 400 episodes each, six canaries) is the first time in the project that a canary arm read its fix: the engine's per-arm `read` count, reproduced exactly by a version-exact join of the sealed traces' causal reads to the platform-authored versions on the fix branch ($17 = 17$, $1 = 1$, $0 = 0$), gives 17 reads in 23 graded canary-arm sessions for the one fix that rewrote an existing key (`repo-readme-ci`, 74 % of graded sessions, P16 met for it), and 0, 0, 1, 1, 1 for the five that wrote a new key. P13″ (median canary-arm reads per run at the floor of 8) is falsified — 1, 1, 8.5, 0.5 — and P16 pooled is 20 of 108 graded sessions, 19.8 %, at the falsification line. That one fix was read and five were not is finding 13.

13. **The loop's own replays discredited the fix before the arm could read it.** (Found 2026-09-30 in the read record of the cells above; fixed in amfs #455, cut `post-replay-evidence`; the confirmatory fleet-10 cells re-run on it, amendment (aa).) Five of the six corrections were *discredited* by outcome evidence within 1–24 minutes of being written, two of them before their canary's first session. The mechanism, traced on `b2b656c5`: the correction was written at 09:00:09Z with confidence 0.78; the loop's three Tier 2 replays of its cases ran over the next minute and all ended in `success` — Tier 2 *passed* — but two of them carried two recorded attempts each, and the attempt boundary of Level 0 does what it is for: each failed attempt charged the entry it had read with a failure. At $n = 0$ one charge takes 0.78 to 0.41; by 09:01:14Z the posterior was under the discredit threshold and `discredited_at` was set, 74 seconds before the canary started. Step 9b of #454 does not pin a discredited entry ("a fix the record has already failed is the avoid list's"), and ranking pins one to -1, so the arm the branch existed to treat was never served it: 0 reads in 74 sessions. Verified in-process against the production rows: `diff_branch` finds the entry, the branch's semantic pool holds it at rank 9 (similarity 0.78), and 9b returns nothing. `2ae2f0e0` went the same way and then *failed* its canary at 09:49Z on an arm that had never read it (2 recurrences in 8 against 0 in 22) — a failing verdict is not read-gated, so the gate rejected a treatment that was never applied. The one fix that was read (`repo-readme-ci`) was discredited after 23 minutes of live reads and read never again; the one that was not discredited (`f5241329`) was written six minutes before its cell ended. Before finding 12's fix, none of the 30 development and fleet-1 canary corrections had ever been discredited — nobody could read them — so making the fix readable is what exposed the seam. The same replays had written 425 `lesson-contrast-replay-*` entries onto `main` since 2026-09-26, each naming a correction under test as the entry to avoid, for every agent on the account. The engine already held the rule that a replay is not traffic — it is graded by the tier that asked for it, does not vote in a canary and does not count as exposure (finding 8, #797) — but only where the loop grades; the memory still learned from it. #455 completes the rule on the memory side: a replay's outcome writes no outcome row (no evidence on the entries it read, no local evidence, no action prior, no contrast lesson), and a discredited entry is pinned on its own branch all the same, with its evidence on the row and out of the avoid list, because the canary is the test of the fix and can only test what the arm read. Verified through the live API before the rerun: on each of the three branches the correction is the first hit, pinned, marked discredited; on `main` nothing is pinned. What this finding leaves for the rerun to decide is whether the evidence the canary arm then produces vindicates the correction or discredits it for a reason — which is the question the canary exists to answer, and §5 reports it doing both.

The canary inside a cell (see also §4.4). Separately from the defects, no canary reached a decision inside any 100-episode cell (K1–K4 in the readouts): a situated canary grades only in-situation runs, and at `pct`=50 with a floor of 8 per arm it needs 16 in-situation graded sessions after it starts, which a cell with 4–10 post-change exposures of one class cannot supply. This is a property of the protocol, not a defect; the flagship long-horizon experiment (200 episodes, fleet 10) is sized for it, and Table 4's "− canary" row is the reading of what ships when the gate is the replay alone.

P	Where	Prediction (dev seeds, set C unless stated; "conf" = confirmatory seeds)	Verdict
P1	(k) 5k	a shift-drafted fix is graded on runs that read the entry	falsified 3/3
P2	(l) c1	− canary ships a regime fix in every cell; recurrence falls	not met (1/4)
P3	(l) c1	SenseLab's canary reaches a decision in-cell	not met (K1–K4)
P4	(l) 5A	maintainer oracle within 3 pts of outcome learning, set A	+2.5 / +4.2
P5	(l)	judges-only recovers ≥ 0.7 of oracle recovery	pending
P6	(o) c2	every Tier 1-passed regime fix gets a Tier 2 verdict	met (ablation); falsified by seam 7 (SenseLab)
P7	(o) c2	situation recall ≥ 0.80, purity ≥ 0.90	0.71 / 0.92
P7′	(r) c3	embedding stage: recall ≥ 0.80, purity ≥ 0.90	falsified 0.40 / 0.67
P8	(q) c3	replays during a canary are re-stamped and graded	met (26/26)
P9(a,c)	(q) c3	proof set of 3–6 cases; kept-case failures occur	met
P9(b)	(q) c3	$\leq$ 25 % of proof-passing regime fixes non-conditional	falsified 83 %
P10	(s) c5	no fix drafted from a replay; no alias from a replay	met (0 of 47 manifest runs; 134 replays present)
P11	(t) c5	non-conditional share $<$ 50 %; a `ci-fix` kept-case failure; ships fall	(a) falsified 4/4; (b) met, 3; (c) met, 11 → 2
P12	(v) c6	≥ 1 in 2 canaries reach the graded floor on both arms; graded/exposed ≥ 0.8	not met: 4 of 9; graded/exposed 0.12–0.55 (from 0–0.27)
P13	(v) c6	≥ 1 in 3 canaries decide	falsified: 0 of 9 (7 from Tier 1 pass, 2 from inconclusive); 3 of 4 at the floor had 0 recurrence in both arms
P14	(v) c6	no pass harms the kept class by > 10 pp (trace-joined)	undefined: no verdict
P11(a′)	(v) c6	$\geq$ 50 % of kept-case regime drafts state a condition; kept-case rejections < 5 of 23	met, 21 of 37 (author-coded, (y)); CI 11/14 all on the true variable, support 9/23 none on it; rejections 4 of 17 and 11 of 39 — the count fell, the share did not
(x) primary	conf f1	SenseLab − each baseline > 0 on the moved half; exact sign-flip over 16 cells, Holm across the five	met in sign and in cells won: +19/+4/+13/+9/+13 pp, 11/8/11/9/10 of 16; p 0.024/0.544/0.040/0.119/0.040, Holm 0.12/0.54/0.16/0.24/0.16 — none survives
(x) secondary	conf f1	kept half and unchanged classes, same test, Holm within endpoint	kept: +38 vs Frontier (Holm 0.007), −24 vs the context window (1 of 16 cells, Holm 0.001), level with the pgvector arms; unchanged: +23/+9/+16 vs Frontier/rerank/DIY (Holm 0.002/0.035/0.004), +3 and +4 vs the context window and pgvector (Holm 0.13, 0.06)
P11(a′)	conf f1	$\geq$ 50 % of kept-case regime drafts state a condition	met, 27 of 39 (author-coded, same sheet and key); `ci-fix` 20 of 28, 17 of them on the PR's files; `support` 7 of 11, none on the true variable
P12–P14, P13′, P13″, P15′, P16	(z″) e2b, finding 13	as (v), (z), (z′), (z″) on the fleet-10 cells run on `post-canary-visible`	measured, not the confirmatory read (five of six corrections discredited by replays before the arm read them): P12 s59 2 of 2 and s73 1 of 2 at the floor; P13 1 of 6 — a failing verdict on an arm that never read the fix; P14 undefined; P13″ falsified, 1/1/8.5/0.5; P15′ no population; P16 20 of 108 graded sessions (19.8 %), 74 % on the one fix that was read
same	conf e2b-r2	as above, on `post-replay-evidence` (amendment (aa)); the confirmatory fleet-10 read	P12 met, 4 of 6 at the floor on both arms (graded/exposed 0.23–0.36 on five, 1.0 on one: second clause not met); P13′ not met, 1 of 6 decided (failed, zero-count bound, arm read the fix in 5 of 10 graded sessions — below the read floor, so untreated by the class rule); P14/P15/P15′ no population (0 passes; treated kept-class comparisons all under 8 sessions); P13″ met, reads at close 2/5/8/12/14/18, median 10 vs floor 8; P16 met, 57 of 76 graded canary-arm sessions (75 %), control 0 of 54; 0 shipped, 0 reverted

7 Pending, and what we expect

The headline tables are the run on the current cut, `post-canary-ni` (findings 8–11 live; `rsi-c6`). The two predictions preregistered on the cut before it were read on that cut's run (`rsi-c5`) and stand as follows. **P10** (no fix has a manifest run that is a replay; no situation alias is created by a replay) is met: of the 47 runs cited by the 33 fix manifests drafted since the cut, none carries a replay tag, while 134 of the 928 sessions sealed in the same window were replays; and the seal hook sets a replay aside before situation identity runs, so no alias can come from one. **P11** had three clauses. (a) The share of proof-passing regime fixes that do not encode the conditional falls below 50 % from 83 % — *falsified*: 4 of 4. (b) At least one regime fix fails Tier 2 on a kept case on `ci-fix` — met, 3. (c) The number of ships falls — met, 11 to 2 in the ablation, with SenseLab's Tier 2 passes falling from 15 to 3. The falsification of (a) is informative in the way the preregistration anticipated: on that cut the proof removed 19 of 23 regime fixes, and the four survivors were wholesale flips whose kept-half replays passed because the replaying agent did not follow the entry. Where the conditional is lost is therefore now located — in the repair agent's draft, not in the proof — and the two remaining instruments are the live canary, which grades what agents actually do with the entry, and the followed-memory criterion, which separates a wrong rule from a wrong rule that was followed. The drafting change that followed (amendment (v): state the condition that separates the failing runs from the passing ones, with the passing runs in hand) is live on the current cut, and P11(a′) reads it: of the regime fixes whose proof set holds a kept case, at least half state a condition. Coded by the author (amendment (y)) over the 37 such fixes of `rsi-c6` and `rsi-e2b-c6`: 21 of 37 — *met*, with the split by class reported in §4.3. The second clause — kept-case proof rejections fall below 5 of 23 — reads 4 of 17 on this cut and 11 of 39 on the fleet-10 rerun: fewer in number where the run is the same size, the same or a higher share.

P12–P14 are read on the fleet-10 rerun (`rsi-e2b-c6`, §4.4). P12: 4 of 9 canaries reached the graded floor on both arms (not met at ≥ 1 in 2; not falsified), and no canary graded 80 % of the runs that read its entry (0.12–0.55; before the correction 0–0.27). P13: 0 of 9 decided — *falsified*. P14: undefined, no verdict. The falsification of P13 is the informative one. Three of the four canaries at the floor had zero recurrence in both arms: the harm gate had nothing to gate because the fleet had stopped making the mistake before the gate could measure it. Whether a fix for a failure that no longer recurs should ship anyway — as a written condition for the next agent, rather than a live correction — was an engine question we put to the owner; finding 12 answered it differently: the failure had stopped recurring in two arms neither of which had received the fix, so the question did not arise. Why a canary graded only a third to a half of the runs stamped to its arm is the same finding: the arm's runs read the branch's other keys (14–197 reads per canary), never the fix's, so "exposure" counted routing, not treatment. The confirmatory fleet-10 cells were run on the `post-canary-visible` cut with P12–P14 and the read-count predictions of amendments (z), (z′) and (z″) as pre-specified; that run read the fix on one canary arm and found the other five discredited by the loop's own replays before their arms could read them (finding 13), and is reported as measured in Table 19. The confirmatory read of P12–P14 and P13′/P13″/P15′/P16 is the rerun on `post-replay-evidence` (amendment (aa), `rsi-conf-e2b-r2`), reported in §5: the arm reads its fix (P16 75 %, P13″ met), four of six canaries reach the floor on both arms (P12 met), one decides (P13′ not met), nothing ships (P14, P15 without population).

The confirmatory scope, reduced before it ran. Amendment (w), declared before any confirmatory-seed cell and before the analysis freeze, reduces the confirmatory grid to what the paper's claims rest on: seeds 37/41/59/73 on the two set-C families (`support`, `ci-fix`), both agent models (GPT-5.5 and Claude Opus 5), fleet 1 on the C protocol with every system in Table 1 and

both ablations, and fleet 10 on the E2b protocol (200 episodes, change at 50) with SenseLab and its two gate ablations — 22,400 episodes on the `post-canary-ni` cut. Amendment (y), declared the morning the runs launched and before any confirmatory cell, cut the fleet-10 part to SenseLab alone on GPT-5.5 (1,600 episodes; 14,400 in all) after the second development fleet-10 run read the ablations the same way as the first, and replaced the two blind coders with the author (§4.3); the fleet-10 paired ablation comparison therefore has no confirmatory read and is labelled development-only wherever it appears. The fleet-1 cells ran once, on `post-canary-ni`. The fleet-10 cells ran twice: on `post-canary-visible` after finding 12, and again on `post-replay-evidence` after that run exposed finding 13 (amendment (aa), declared before the rerun launched); both are reported, the first as measured and the second as the confirmatory read, 16,000 confirmatory episodes in all. The loop has not been run on the remaining eight families: they appear only in the level-0 comparison of the earlier ten-family grid (SenseLab with the loop held, against the same baselines, change sets A and B, development seeds), cited as such; their held-out set C is not authored, and the held-out-family design of amendment (l) lapses with it. The paired test is the exact sign-flip test over cells with Holm across systems (amendment (x)); the mixed-effects learning-curve model named in the preregistration is not run, for the reasons given there. What is not run at all and is reported as not run: verdict sources (E3), the public track (E4), the A/A, adversarial, reward-hacking and manifest-calibration probes, and the Managed Model rungs above L0 (E7). Each is a claim this paper does not make.

8 Related work

Self-improvement. Self-Refine [9] and STaR [10] improve a model's outputs or weights from its own feedback within a task or a training run. The Gödel machine lineage [11, 12] and AlphaEvolve [13] modify the improver itself, gated by a proof or a benchmark score, in a sandbox. The engine here is the deployed case: the object it modifies is what every live agent acts on, the gate is live traffic rather than a held-out score, and the discipline — manifest, tiers of proof, staged exposure, reversibility, an audit trail — is the contribution rather than the search. The "fixed budget, held-out, sustained, generalising" conditions that the preregistration adopts for any claim of self-improvement (E5) are the four conditions Weco proposes for the "net positive" rung of its ladder of recursive self-improvement [17] and reports meeting with AIDE^2 [18]: a fair human baseline, a sustained multi-step trend, generalisation beyond the optimised measurement, and a fixed budget. (The ladder's rungs number the strength of the claim; the levels of this paper number the stages of one loop. The two numberings are unrelated.) E5 has not run (§7); no rung is claimed.

Release engineering. Canary release and controlled online experiments [14] are the discipline the loop borrows for level 1; A/A tests and look-schedule inflation are the standard hazards, and the preregistration's A/A probe measures the empirical false-pass rate of the canary's test under the schedule the harness uses.

Benchmarks. τ-bench and τ^2-bench [15, 16] evaluate tool-using agents against a policy that stays fixed across tasks; our public track (E4) amends the policy between passes to ask this paper's question on a public task set. Continual-learning benchmarks for language models measure forgetting in weights; the object here is what a deployed agent acts on, and the failure mode is remembering.

Agent memory, as substrate and as baseline. Reflexion [2] and Voyager [3] write verbal reflections and skills back for later episodes; Generative Agents [4] and MemGPT [5] manage a store with retrieval and summarisation; Mem0 [6] and Zep [7] are production memory layers evaluated on conversational recall (LoCoMo, LongMemEval [8]). All are evaluated on retrieving what was stored; none tests, gates or reverts a change to the store, and none is measured on a change of the world after storage. They are the substrate a self-improving engine operates on and the baselines it is measured against here; our earlier report [1] measured level 0 against these designs under a regime change.

9 Threats to validity

- **Development seeds.** Every number is from two seeds on two families, chosen for development. The confirmatory seeds run once after the freeze; nothing here is a confirmed difference, and the intervals say so.
- **One model, one effort level.** GPT-5.5 at low effort. Claude Opus 5 has not run on set C on the development seeds; it is half of the confirmatory grid (amendment (w)), where every cell is model-aware.
- **The benchmark's outcome is coarse.** The harness reports `success` for an episode solved within three attempts; that is what the customer reports, and it hid finding 9 from the proof set for two runs. The clean-success rule reads the attempts the agent already records; a customer whose agent never records attempts sees no change and gets no protection.
- **The engine changed between runs.** Eleven defects were fixed in production during the study (amfs-internal #780–#802, OSS #454–#455), one of them inside the fleet-1 confirmatory window and two between the two fleet-10 confirmatory runs. Every result carries its cut; latency claims are within one cut only; and the changes are to the loop's evidence path, not to any threshold constant, which the preregistration forbids. The one gate rule that changed — the canary's primary, from superiority to non-inferiority (finding 11) — was preregistered with its falsifier before the run that tests it.
- **The loop's drafting cost is an estimate.** Judge verdicts and Tier 2 replays are charged exactly per cell; the repair agent's spend is the account ledger's remainder spread over every draft on the account (§4.6), so a cell that drafted long fixes is under-charged and one that drafted short ones over-charged. The confirmatory runs snapshot the ledger per cell and report the difference directly.
- **Hand-coding.** The "conditional" share (P9(b), P11, P11(a′)) is a reading of the draft text against written instructions by one coder, the author, who is not blind to arm or verdict (amendment (y)). Every draft, the instructions, the coding and the key are published so the reading can be redone; the class split in §4.3 is the part a second coder is least likely to move.
- **The harness routes the canary.** The production router runs in the MCP gateway; the harness applies the same hash itself (parity-tested in CI) because it talks to the HTTP API.
- **The context window is a strong baseline where it fits.** On `ci-fix` the whole cell fits in 24k tokens and the window sees the change; it is the right comparison for short horizons and the wrong one for the 200-episode fleet, where it cannot fit.

- **We built the system.** The preregistration, the amendments with their falsifiers, the cut records and the ledger exist so that a reader can check what was predicted before each run rather than take our reading of it.

A Reproducibility

Runs. Harness arm names, cell ids and command lines: baselines `rsi-c1-tier1` (`none`, `pgvector+rerank`, `senselab`), `rsi-c1-baselines` (`pgvector`, `pgvector-diy+outcomes`), `rsi-c1-baselines-cw` (`raw-traces`); SenseLab `rsi-c1-rsi`, `rsi-c2`, `rsi-c3`, `rsi-c5`, `rsi-c6` (`senselab-rsi`, `senselab-rsi-nocanary`); fleet 10 `rsi-e2b-c5`, `rsi-e2b-c6` (plus `senselab-rsi-noreplay`, 200 episodes, change at 50, `-rep`); confirmatory `rsi-conf-f1-s<seed>`, `rsi-conf-e2b-s<seed>` (the `post-canary-visible` run, finding 13) and `rsi-conf-e2b-r2-s<seed>` (the `post-replay-evidence` rerun; `CL_CONF_RUN_TAG=r2`) from `confirm_rsi.sh`. Cell id `<family>|<arm>|gpt-5.5|s<seed>|e100-d300-n0.0-f0-a1-c30-x0.5-oid-vC`. Command:

```
python -m continual_learning.runner --run <run> --scenarios support ci-fix --arms <arms> --models
gpt-5.5 --seeds 11 23 --fleet 1 --episodes 100 --change-at 30 --change-focus 0.5 --change-variant
C --opaque-ids --workers 2
```

with `CL_RSI_TIER2=1` and a Tier 2 replay receiver reachable from the service.

Engine cuts. `pre-seam4`: amfs-internal #787/#788. `post-seam5`: #793 (07:00:22Z), #794 (08:27:24Z, 2026-09-27). `post-seam5b`: #795 (12:50:24Z), #796 (13:39:13Z). `post-seam8`: #797 (09:14:17Z, 2026-09-28). `post-clean-success`: #798. `post-canary-ni`: #799 (11:35:32Z, 2026-09-29). `post-canary-zero`: OSS #454, amfs-internal #800/#801 (03:40:19Z, 2026-09-30). `post-canary-visible`: #802 (08:21:05Z, 2026-09-30). `post-replay-evidence`: OSS #455 (12:52:04Z, 2026-09-30; the confirmatory fleet-10 cut). Each cut's Cloud Run revision is in `HYPOTHESIS_RSI.md`.

Analysis. `python -m benchmarks.continual_learning.analysis.rsi --paper` writes `results/rsi-report/REPORT.md` and `report.json` for the development seeds; `--seeds 37 41 59 73 --models gpt-5.5 claude-opus-5 --runs <confirmatory runs> --report-dir results/rsi-report-confirmatory` is the confirmatory read, run on the staged copy of the fleet-1 results (one run per system, §5; the staging script and its byte-identity check are in the package); `analysis.rsi_canary_readout --run <run>` is the per-canary read record (verdicts, reads by class of run, the version-exact join); the three-class canary-arm read count of findings 12–13 is a supplementary script in the package that reads the same tables; `analysis.rsi_coding_sheet make --runs <runs>` and `score --answers` produce and score the P11(a′) sheets; `analysis.fill_rsi_paper` fills this document from the report and `analysis.rsi_figures` draws the figures. `analysis.platform_calibration` reads the account ledger for the cost table (§4.6). The analysis code is tagged `analysis-freeze-rsi` before the confirmatory runs. The preregistration, amendments, readouts (`HEADROOM_RSI.md`), change sets, run records and the coding sheets (drafts, instructions, author's coding, key) are published with the paper.

References

[1] SenseLab Research. From memory to recursive self-improvement: agents that learn from outcomes, agents that only remember, and a model trained on what they learned, when the tasks never repeat and the rules change. Preprint, 2026.

[2] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, S. Yao. Reflexion: Language agents with verbal reinforcement learning. NeurIPS 2023.

[3] G. Wang et al. Voyager: An open-ended embodied agent with large language models. TMLR 2024 (arXiv:2305.16291).

[4] J. S. Park et al. Generative agents: Interactive simulacra of human behavior. UIST 2023.

[5] C. Packer et al. MemGPT: Towards LLMs as operating systems. arXiv:2310.08560, 2023.

[6] P. Chhikara et al. Mem0: Building production-ready AI agents with scalable long-term memory. arXiv:2504.19413, 2025.

[7] P. Rasmussen et al. Zep: A temporal knowledge graph architecture for agent memory. arXiv:2501.13956, 2025.

[8] D. Wu et al. LongMemEval: Benchmarking chat assistants on long-term interactive memory. ICLR 2025.

[9] A. Madaan et al. Self-Refine: Iterative refinement with self-feedback. NeurIPS 2023.

[10] E. Zelikman, Y. Wu, J. Mu, N. D. Goodman. STaR: Bootstrapping reasoning with reasoning. NeurIPS 2022.

[11] J. Schmidhuber. Gödel machines: Fully self-referential optimal universal self-improvers. In *Artificial General Intelligence*, Springer, 2007.

[12] J. Zhang, S. Hu, C. Lu, R. Lange, J. Clune. Darwin Gödel Machine: Open-ended evolution of self-improving agents. arXiv:2505.22954, 2025.

[13] A. Novikov et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. arXiv:2506.13131, 2025.

[14] R. Kohavi, D. Tang, Y. Xu. *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*. Cambridge University Press, 2020.

[15] S. Yao, N. Shinn, P. Razavi, K. Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. arXiv:2406.12045, 2024.

[16] V. Barres et al. τ^2-bench: Evaluating conversational agents in a dual-control environment. arXiv:2506.07982, 2025.

[17] Weco AI. 4 levels of recursive self-improvement. Blog post, July 2026. `https://www.weco.ai/blog/4-levels-of-recursive-self-improvement`.

[18] Z. Jiang et al. AIDE^2: First evidence of recursive self-improvement. Weco AI blog post, July 2026. `https://www.weco.ai/blog/first-evidence-of-recursive-self-improvement`.